CHAPTER 20
ELECTROMAGNETISM

4.5 Electromagnetic effects

4.5.1 Electromagnetic induction

1 Describe an experiment to demonstrate electromagnetic induction

2 State that the magnitude of an induced e.m.f. is affected by:
   (a) the rate of change of the magnetic field or the rate of cutting of magnetic field lines
   (b) the number of turns in a coil

3 State and use the fact that the effect of the current produced by an induced e.m.f. is to oppose the change producing it (Lenz's law) and describe how this law may be demonstrated

4.5.2 The a.c. generator

1 Describe a simple form of a.c. generator (rotating coil or rotating magnet) and the use of slip rings and brushes where needed

2 Sketch and interpret graphs of e.m.f. against time for simple a.c. generators and relate the position of the generator coil to the peaks, troughs and zeros of the e.m.f.

4.5.3 Magnetic effect of a current

1 Describe the pattern and direction of the magnetic field due to currents in straight wires and in solenoids and state the effect on the magnetic field of changing the magnitude and direction of the current

2 Describe how the magnetic effect of a current is used in relays and loudspeakers and give examples of their application

4.5 Electromagnetic effects continued

4.5.4 Forces on a current-carrying conductor

1 Describe an experiment to show that a force acts on a current-carrying conductor in a magnetic field, including the effect of reversing:

(a) the current

(b) the direction of the field

2 Recall and use the relative directions of force, magnetic field and current

3 Describe the magnetic field patterns between currents in parallel conductors and relate these to the forces on the conductors (excluding the Earth's field)

4.5.5 The d.c. motor

1 Know that a current-carrying coil in a magnetic field may experience a turning effect and that the turning effect is increased by increasing:

(a) the number of turns on the coil

(b) the current

(c) the strength of the magnetic field

2 Describe the operation of an electric motor, including the action of a split-ring commutator and brushes

4.5.6 The transformer

1 Describe the structure and explain the principle of operation of a simple iron-cored transformer

2 Use the terms primary, secondary, step-up and step-down

3 Recall and use the equation

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

where P and S refer to primary and secondary

4 State the advantages of high-voltage transmission and explain why power losses in cables are smaller when the voltage is greater

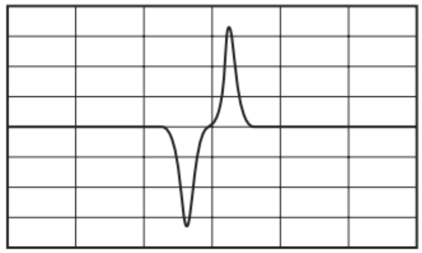
4.6 Uses of an oscilloscope

1 Describe the use of an oscilloscope to display waveforms (the structure of an oscilloscope is **not** required)

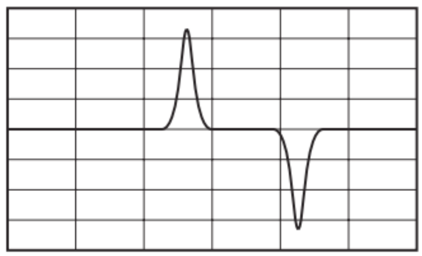
2 Describe how to measure p.d. and short intervals of time with an oscilloscope using the Y-gain and timebase

1. O/N/23/11/Q29

A magnet moves at constant speed through a coil of wire. The coil is connected to an oscilloscope. The diagram shows the trace produced on the screen.



The experiment is changed and a new trace is produced.



Which statement explains why the new trace is different from the original trace?

A The coil is longer and the other pole of the magnet enters the coil first.

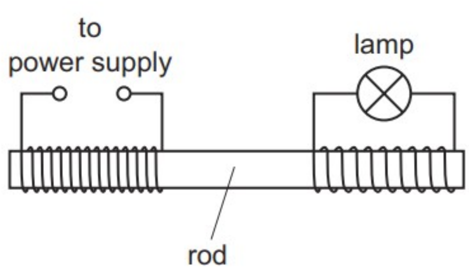
B The coil is longer and the same pole of the magnet enters the coil first.

C The coil is shorter and the other pole of the magnet enters the coil first.

D The coil is shorter and the same pole of the magnet enters the coil first.

2. O/N/23/11/Q34

A girl makes a simple transformer to illuminate a low voltage lamp. The transformer is shown.

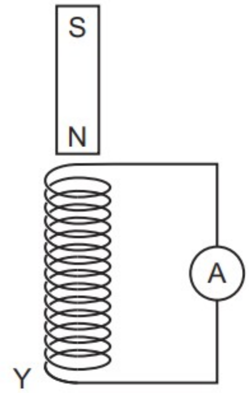


What is the material of the rod and what is the type of power supply?

	material of rod	power supply
A	plastic	d.c. supply
B	plastic	a.c. supply
C	iron	d.c. supply
D	iron	a.c. supply

3. O/N/23/11/Q30

A bar magnet is dropped through a vertical solenoid that is connected to a sensitive ammeter.



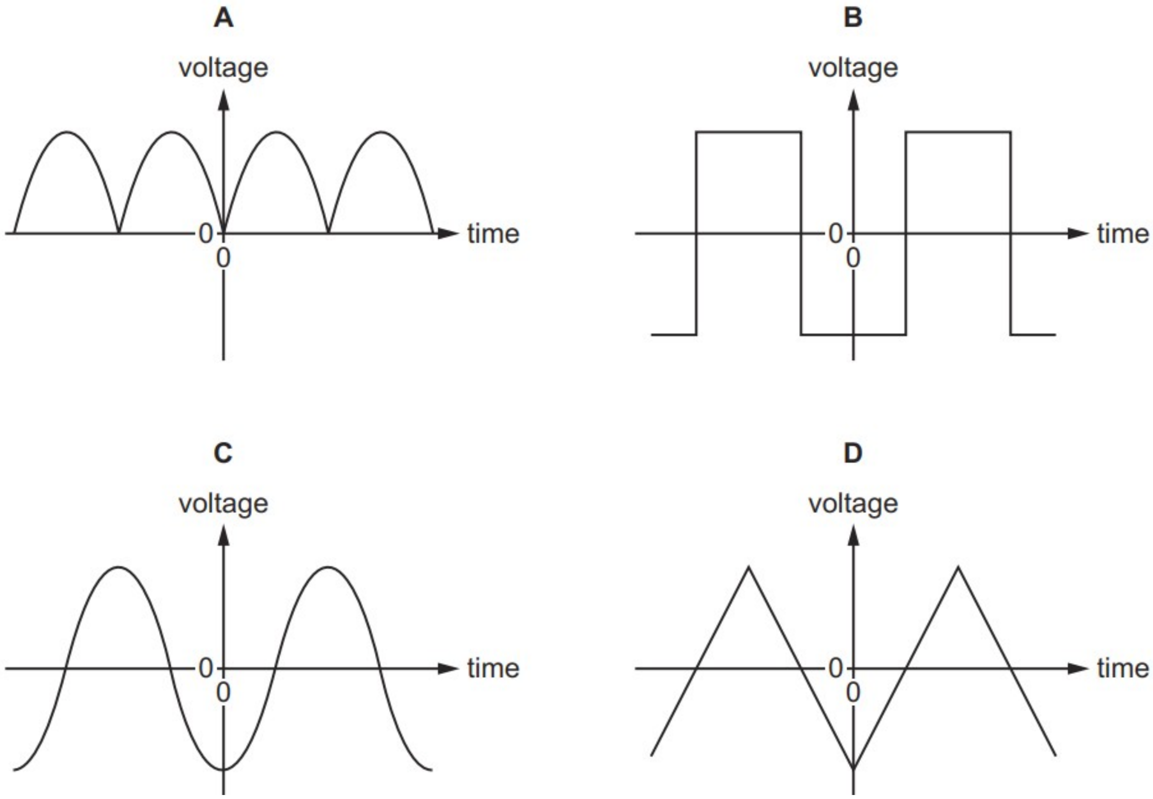
A current is induced in the solenoid and the solenoid becomes an electromagnet.

Which row describes the polarity of end Y of the solenoid and the force between the solenoid and the bar magnet when the bar magnet leaves the solenoid?

	polarity of end Y of the solenoid	force between solenoid and bar magnet
A	north	attraction
B	north	repulsion
C	south	attraction
D	south	repulsion

4. O/N/23/11/Q31

Which graph shows the voltage output against time for a simple a.c. generator?



5. O/N/23/11/Q32

A charger for a mobile phone (cellphone) uses a transformer which steps the voltage down from 230 V to 5.0 V.

The primary coil in the transformer has 2500 turns.

How many turns are on the secondary coil?

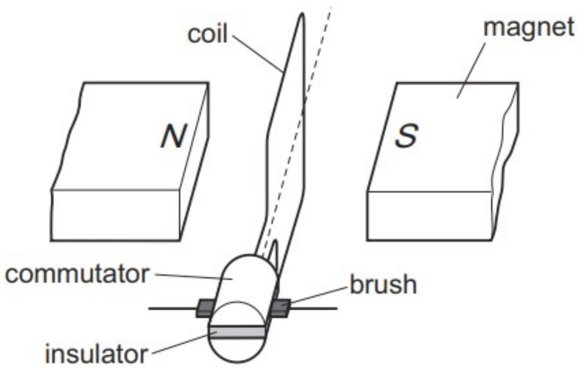
A 11 **B** 54 **C** 500 **D** 11 500

6. O/N/23/11/Q33

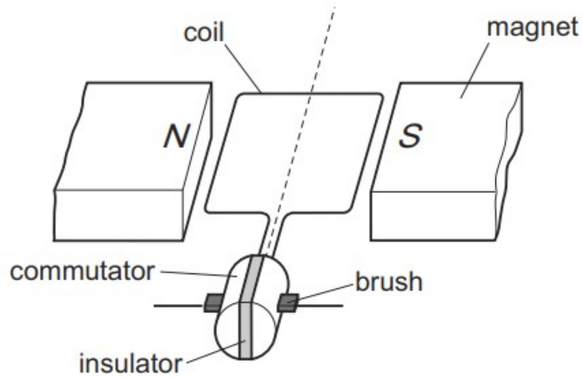
A commutator is connected to the coil of a d.c. motor.

The commutator rotates and connects the coil to the power supply.

Once every half-turn, the current in the coil reverses.



coil perpendicular to magnetic field



coil parallel to magnetic field

What is the name of the commutator and what is the orientation of the plane of the coil as the current reverses?

	name	orientation
A	slip ring	parallel to the magnetic field
B	slip ring	perpendicular to the magnetic field
C	split ring	parallel to the magnetic field
D	split ring	perpendicular to the magnetic field

7. O/N/23/12/Q35

The output of an a.c. generator is connected to a lamp.

The coil of the generator is rotated at a constant rate which makes the lamp flash twice every second.

The coil is then rotated at half the rate of rotation.

What happens to the maximum brightness of the lamp and how many times does the lamp flash every second?

	brightness	number of flashes every second
A	decreases	1
B	decreases	4
C	does not change	1
D	does not change	4

8. O/N/23/12/Q36

Which device uses the magnetic effect of an electric current?

A compass

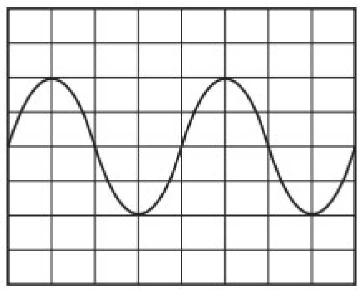
B fuse

C loudspeaker

D magnifying glass

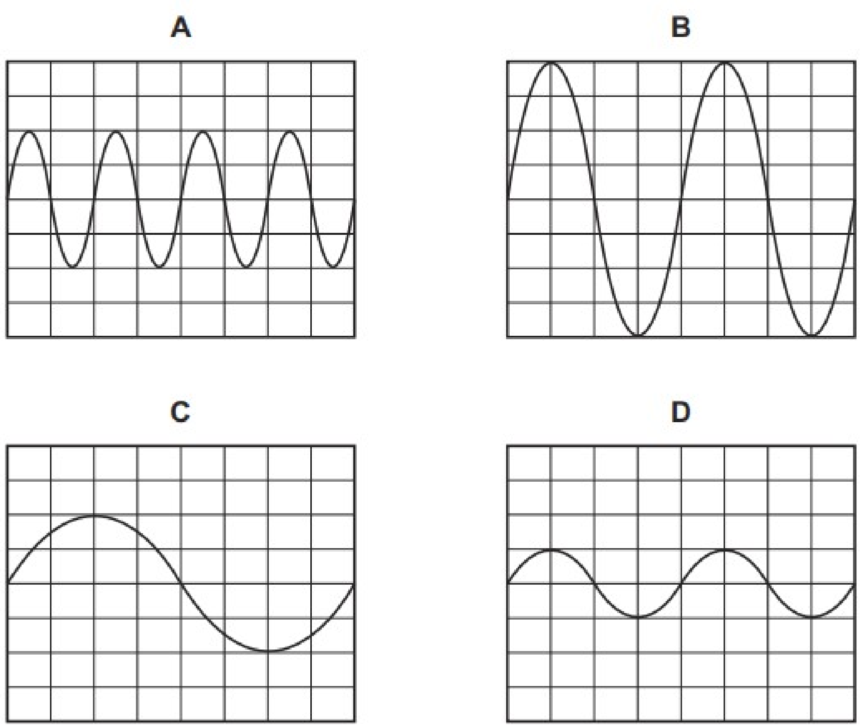
9. O/N/23/12/Q38

The diagram shows the trace produced on the screen of an oscilloscope due to an alternating voltage input.



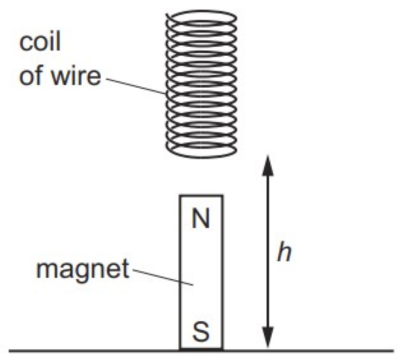
The time–base on the oscilloscope is set to 4 ms / cm.

What does the trace look like when the time base is set to 2 ms / cm?



10. M/J/23/12/Q32

A coil of wire above a magnet is dropped. As the coil falls over the magnet, an e.m.f. is induced across the coil.



What does **not** affect the magnitude of the e.m.f. induced across the coil?

A changing the number of turns of wire in the coil

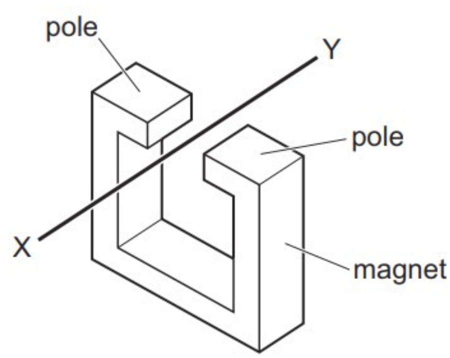
B changing the height h from which the coil is dropped

C changing the direction of the magnet, so the north pole is at the bottom

D changing the strength of the magnetic field from the magnet

11. M/J/23/11/Q32

When wire XY is moved downwards between the poles of a stationary magnet, an e.m.f. is produced across X and Y.



Which action produces an e.m.f. across X and Y in the opposite direction?

A Move both the wire and the magnet upwards at the same speed.

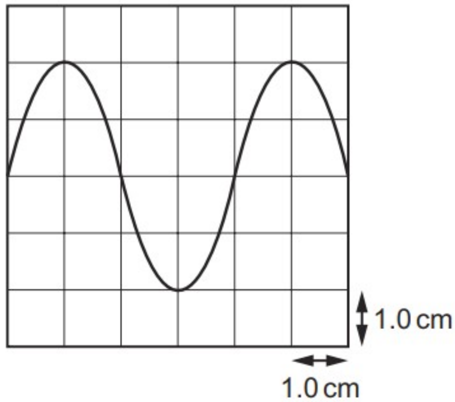
B The wire is kept stationary and the magnet is moved upwards.

C The wire is moved downwards and the magnet is moved upwards.

D The wire is moved upwards and the magnet is kept stationary.

12. M/J/23/11/Q33

An a.c. voltage is displayed on an oscilloscope screen. The Y-gain is set at 2.0 V / cm.



What is the maximum value of the voltage?

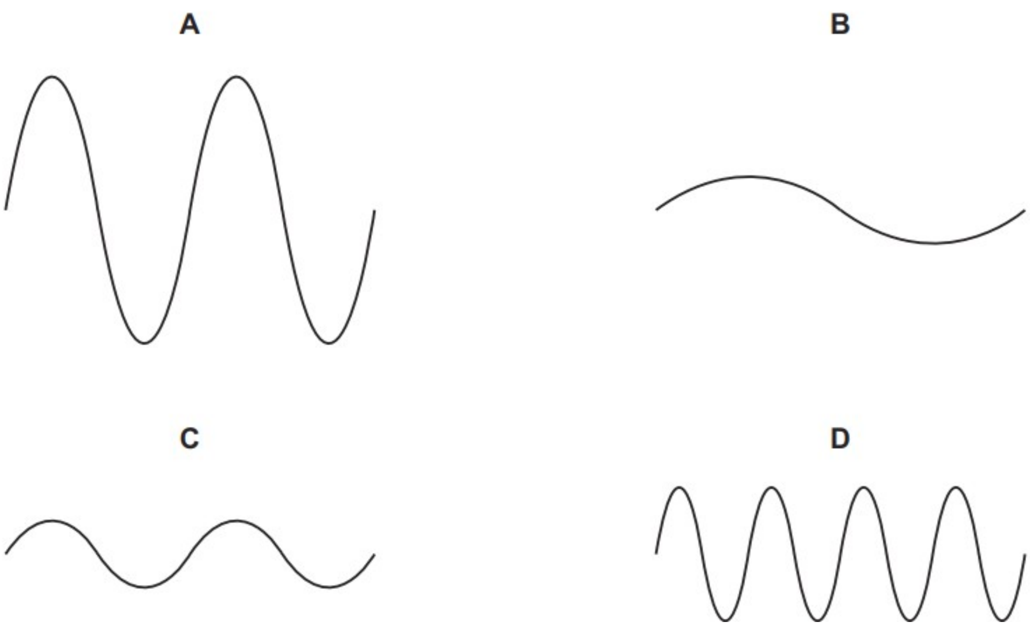
A 2.0 V **B** 4.0 V **C** 8.0 V **D** 12 V

13. O/N/22/11/Q34

The diagram shows the output from an a.c. generator displayed on an oscilloscope screen.

The generator is turned at a slower rate. The settings on the oscilloscope controls are not changed.

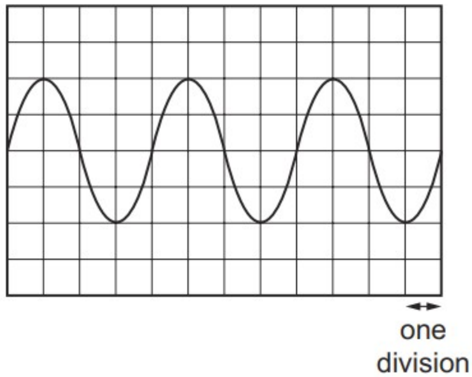
Which diagram shows the new output?



14. O/N/22/11/Q35

The sound wave produced by a note is displayed on the screen of a cathode-ray oscilloscope.

The time-base is set at 5 ms for one division.



What is the frequency of the note?

A 20 Hz **B** 50 Hz **C** 60 Hz **D** 150 Hz

15. O/N/22/11/Q33

Which statement about a transformer is correct?

A The changing magnetic field in the transformer induces an e.m.f. in the secondary coil.

B The core of the transformer is made of iron because iron is a good electrical conductor.

C The transformer converts alternating current to direct current.

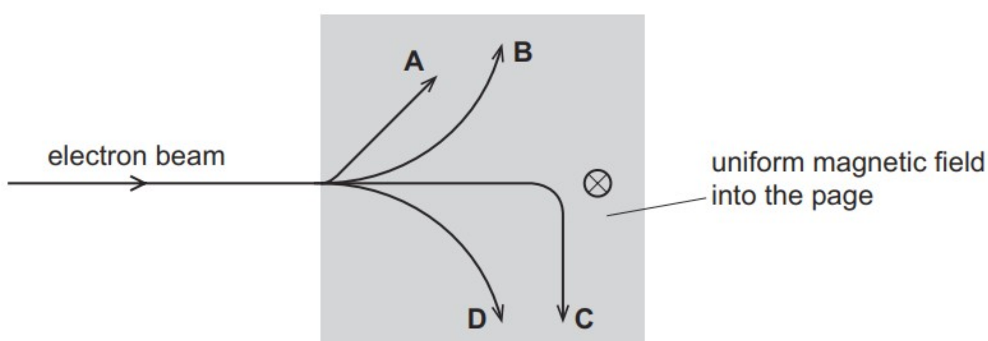
D The transformer converts direct current to alternating current.

16. O/N/22/11/Q37

A beam of electrons is fired into a uniform magnetic field as shown.

The direction of the magnetic field is into the page.

Which path do the electrons follow?



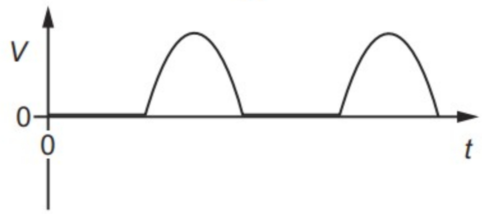
17. O/N/22/12/Q35

A coil of wire is rotated at a constant rate between the poles of a U-shaped magnet.

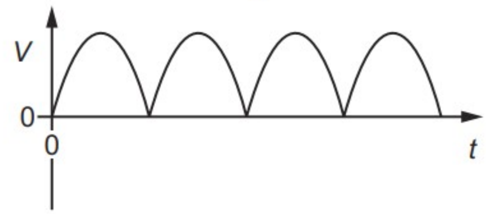
The two ends of the coil are connected to different slip rings.

Which graph shows how the voltage V between the slip rings varies with time t?

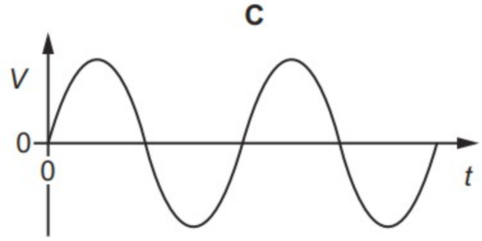
A



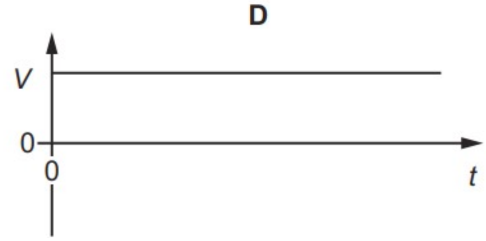
B



C

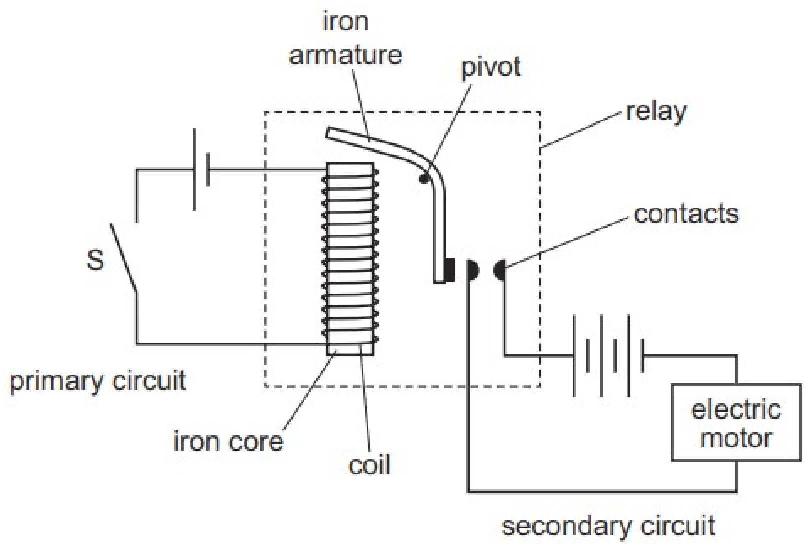


D



18. O/N/22/12/Q38

The diagram shows a relay used to switch on an electric motor.



A student makes five statements to explain how the relay switches on the electric motor. The statements are **not** in the correct order.

1 The current in the coil magnetises the electromagnet.

2 The armature closes the contacts.

3 The current in the secondary circuit makes the motor turn.

4 The electromagnet attracts the iron armature.

5 The switch S in the primary circuit is closed.

What is the correct order of the statements?

A $1 \rightarrow 5 \rightarrow 2 \rightarrow 4 \rightarrow 3$

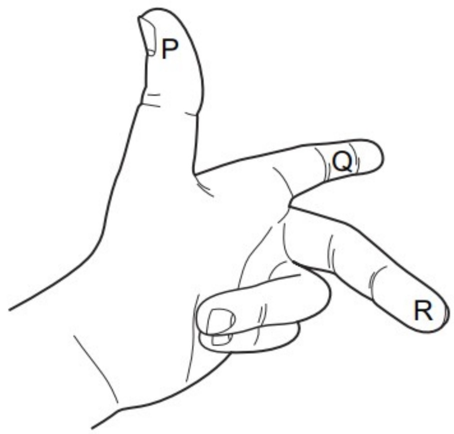
B $3 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 5$

C $3 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 2$

D $5 \rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 3$

19. M/J/22/12/Q38

A left hand can be used to determine the direction of the force when a current-carrying conductor is perpendicular to a magnetic field.

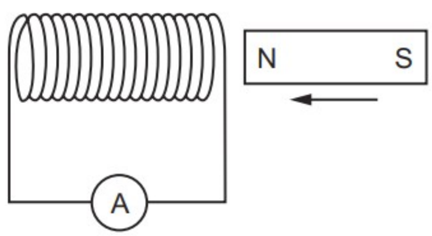


Which quantities are represented by the direction of fingers P, Q and R?

	P	Q	R
A	current	field	force
B	field	force	current
C	force	current	field
D	force	field	current

20. M/J/22/12/Q39

As a magnet is moved into the coil of wire as shown, there is a small positive reading on the sensitive ammeter.



Which change must increase the size of the reading?

A moving the opposite pole into the coil

B pulling the magnet out of the coil

C pushing the magnet in faster

D unwinding some of the turns of wire

21. M/J/22/12/Q40

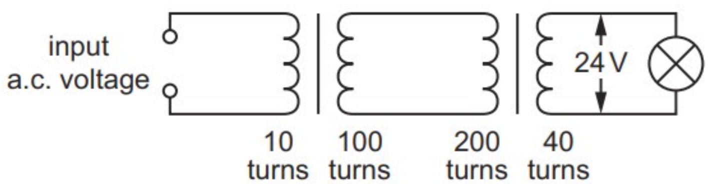
Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

22. M/J/22/11/Q37

The diagram shows the circuit for a model power line with two transformers and a lamp at the output.

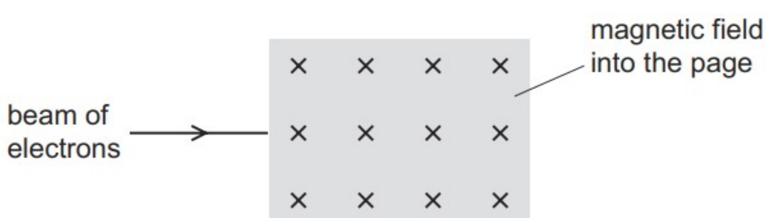


What is the input voltage?

A 0.12 V **B** 6.0 V **C** 12 V **D** 96 V

23. M/J/22/11/Q39

The diagram shows a beam of electrons about to enter a magnetic field. The magnetic field is directed into the page.



What is the direction of the deflection of the electrons as they enter the magnetic field?

A into the page

B out of the page

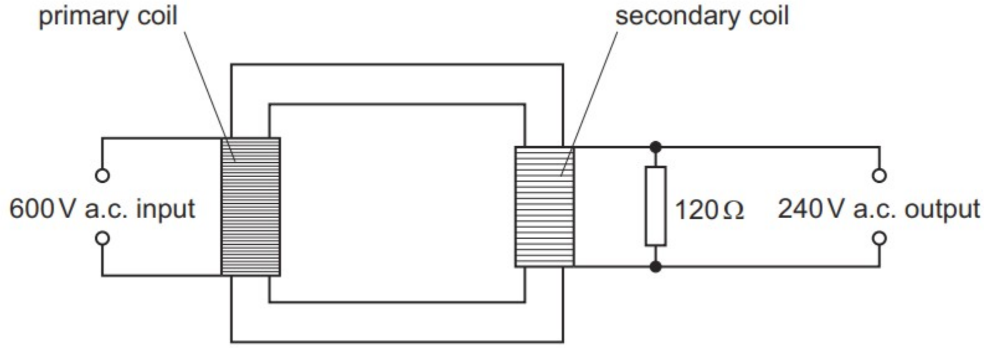
C up the page

D down the page

24. O/N/21/11/Q34

A transformer with an efficiency of 100% has a primary voltage input of 600 V and a secondary voltage output of 240 V.

The secondary coil is attached to a resistor of resistance 120 Ω.

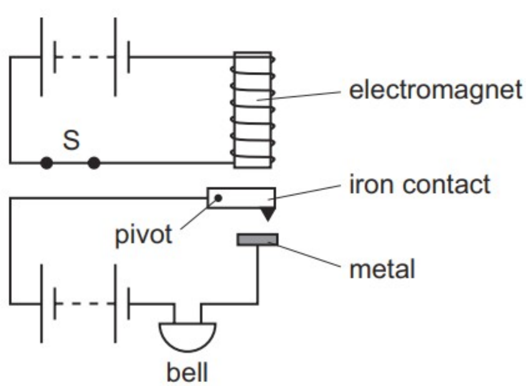


What is the power dissipated in the resistor and what is the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

25. O/N/21/11/Q35

The diagram shows an alarm system in which the switch S is shown closed. The top circuit is arranged so that the electromagnet is positioned over the soft iron contact.

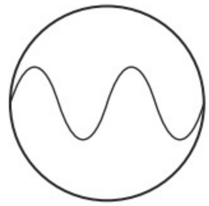


What happens when the switch S is opened?

	iron contact	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

26. O/N/21/11/Q36

A microphone detects a musical sound and the signal is fed into an oscilloscope. The diagram shows the trace which appears on the screen.



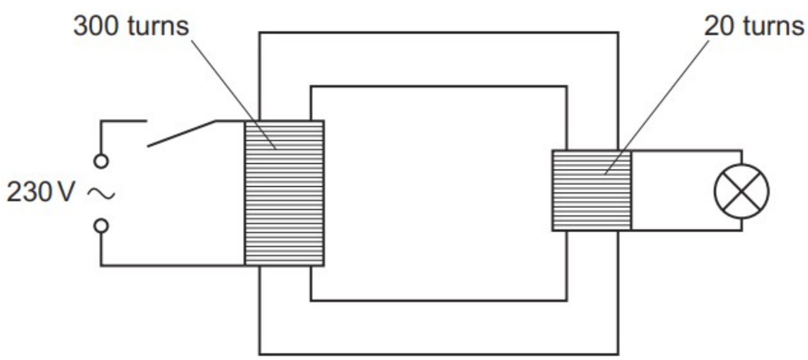
The spot of light on the oscilloscope screen takes 1.2 ms to travel across the screen.

What is the frequency of the musical sound?

A 170 Hz **B** 330 Hz **C** 1700 Hz **D** 3300 Hz

27. M/J/21/12/Q38

A student uses a transformer to light a filament lamp using a 230 V a.c. supply. The lamp has a maximum voltage rating of 6.0 V.



What happens when the circuit is switched on?

A The lamp does not light at all.

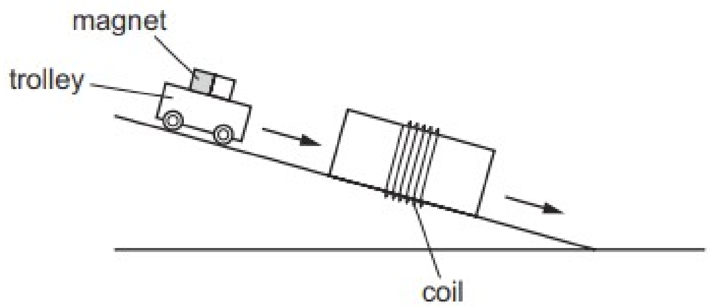
B The lamp lights at normal brightness.

C The lamp lights dimly.

D The lamp lights up brightly and then goes out.

28. M/J/21/12/Q37

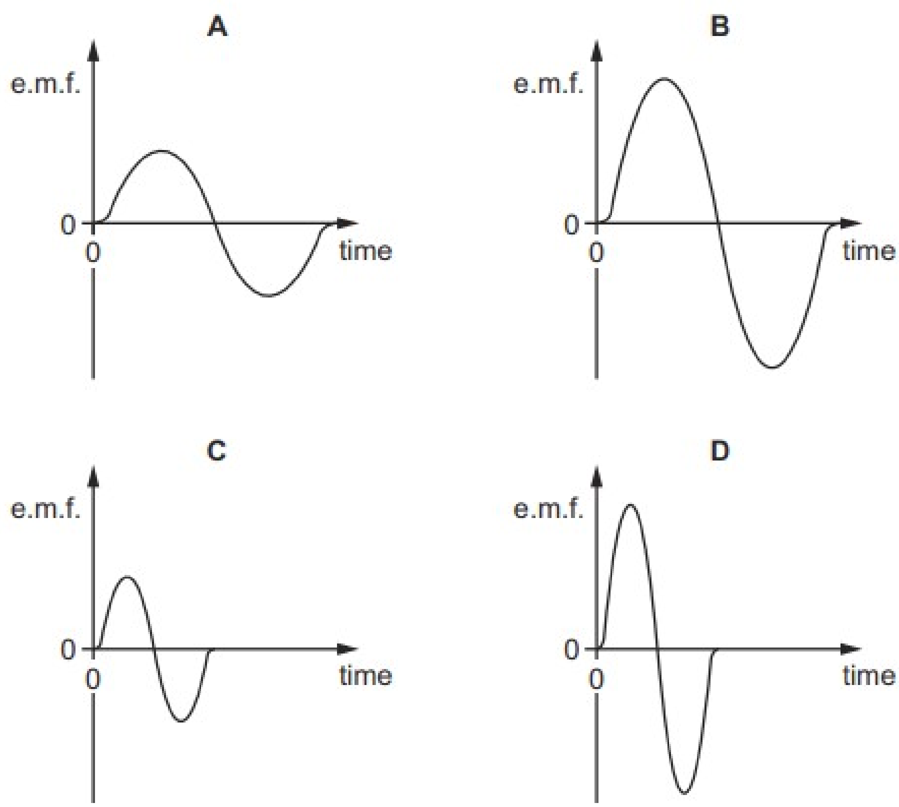
A trolley carrying a strong magnet rolls down a ramp at constant speed. It passes through a coil as shown.



An electromotive force (e.m.f.) is induced in the coil. A graph of the e.m.f. against time is plotted.

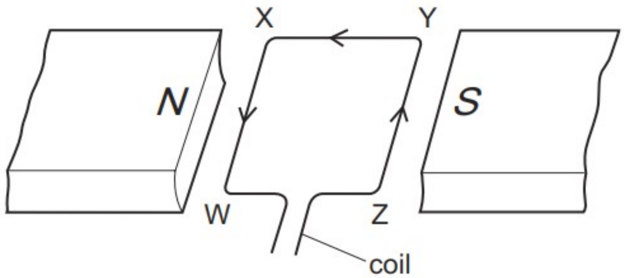
The experiment is repeated with different coils and with a steeper ramp. The trolley moves at a greater constant speed on the steeper ramp.

Which graph is produced using the coil with the least number of turns and the steepest ramp? All graphs are drawn to the same scale.



29. M/J/21/11/Q32

The diagram shows a horizontal rectangular wire coil WXYZ between the poles of a magnet.



There is a current in the coil in the direction shown.

Which statement is correct?

A The side WX experiences an upward force.

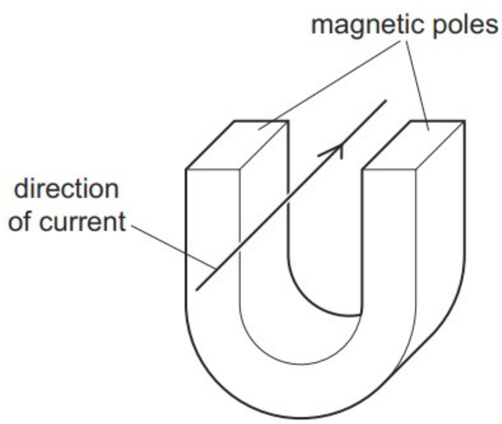
B The side XY experiences an outward force.

C The side YZ experiences an inward force.

D The side ZW experiences a downward force.

30. M/J/21/11/Q34

The diagram shows a current-carrying conductor between the poles of a magnet. The force on the wire acts downwards.



Four changes are possible.

1 The current is increased.

2 A stronger magnet is used.

3 The current is reversed.

4 The poles exchange positions.

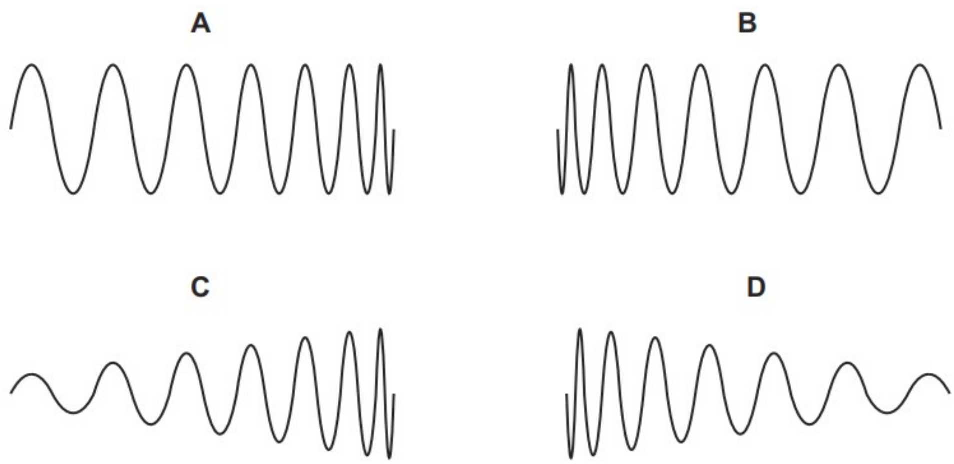
Which two changes made together keep the force acting downwards?

A 1 and 3 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

31. M/J/21/11/Q35

In an alternating current (a.c.) generator, a magnet rotates near a coil of wire. The induced electromotive force (e.m.f.) in the coil is displayed on an oscilloscope screen.

Which trace is produced as the magnet slows down?



32. O/N/20/12/Q34

What is used with a magnet to create an induced electromotive force (e.m.f.) in a simple a.c. generator?

A a battery

B a coil of wire

C a voltmeter

D a relay

33. O/N/20/12/Q35

What changes kinetic (movement) energy into electrical energy?

A a car battery

B an electric fire

C an electric motor

D a wind turbine

34. O/N/20/12/Q36

A transformer has 4800 turns on its primary coil and 480 turns on its secondary coil.

The primary coil is connected to a 240 V a.c. supply. The secondary coil is connected to a lamp.

How does the output current in the lamp compare with the input current?

A higher frequency a.c.

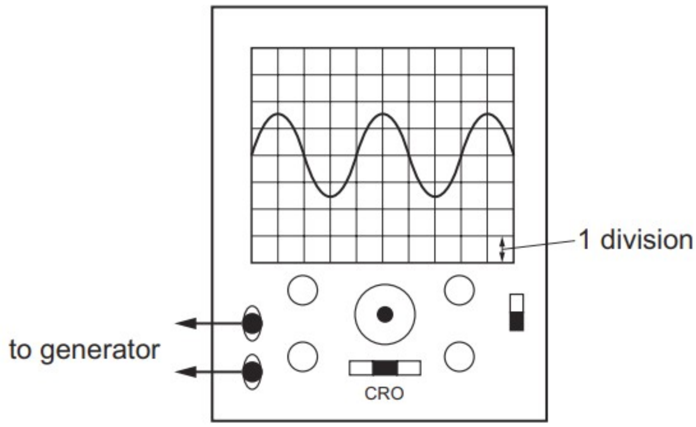
B lower frequency a.c.

C current in one direction only

D the same frequency a.c.

35. O/N/20/12/Q37

A cathode-ray oscilloscope with the Y-gain set at 2 V / division is connected to a generator.



Which row describes the signal from the generator and its amplitude?

	signal	amplitude / V
A	a.c.	3
B	a.c.	6
C	d.c.	3
D	d.c.	6

36. M/J/20/12/Q39

A rectangular current-carrying coil is pivoted between the poles of an electromagnet.

Which action does **not**, on its own, increase the size of the turning effect exerted on the coil?

A increasing the current in the coils of the electromagnet

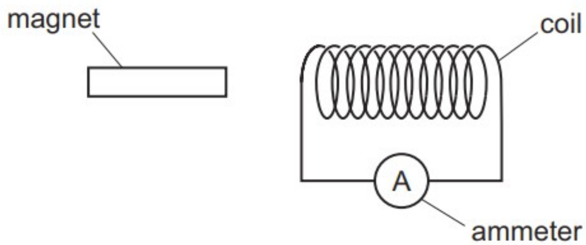
B increasing the current in the rectangular coil

C reversing the current in the electromagnet

D increasing the number of turns on the rectangular coil

37. M/J/20/12/Q40

The diagram shows a coil connected to a very sensitive ammeter. A magnet is next to the coil.



Which action results in a zero reading on the ammeter?

A moving the coil and the magnet at the same speed in opposite directions

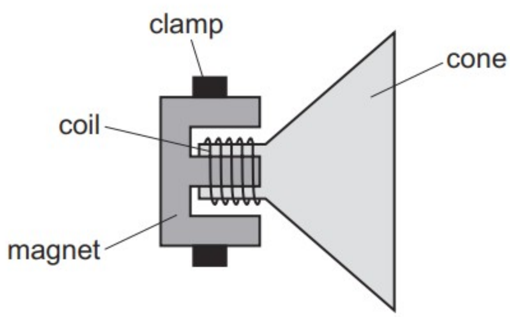
B moving the coil and the magnet at the same speed in the same direction

C moving the coil away from the stationary magnet

D moving the magnet towards the stationary coil

38. M/J/20/11/Q33

There is varying current in the coil of the loudspeaker shown. The loudspeaker is producing a sound. The magnet is clamped.



What is vibrating to produce the sound?

A coil only

B cone only

C magnet only

D coil and cone

39. O/N/19/11/Q37

The input voltage to a transformer is 24 V a.c. and the output voltage is 6.0 V.

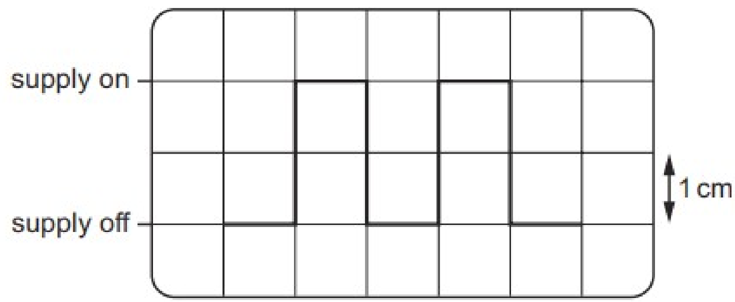
The input coil has 720 turns.

How many turns are on the output coil?

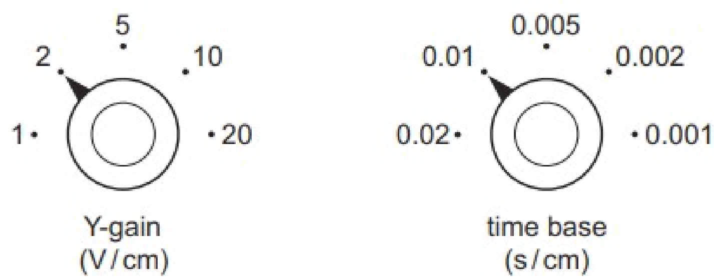
A 5 **B** 180 **C** 2900 **D** 100 000

40. O/N/19/11/Q39

The diagram shows the trace on an oscilloscope screen when connected to a d.c. supply which is being switched on and off.



The settings of the Y-gain control and the time base control are shown.



The voltage of the supply is *v*.

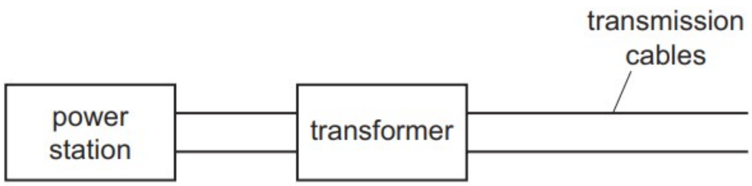
The supply is switched on *f* times per second.

What are the values of *v* and of *f*?

	v / V	*f*
A	2	50
B	2	100
C	4	50
D	4	100

41. O/N/19/12/Q37

Transformers are used to transmit electrical energy between power stations and transmission cables, as shown.



What is the purpose of the transformer in the diagram?

A to decrease the current and the potential difference from the power station

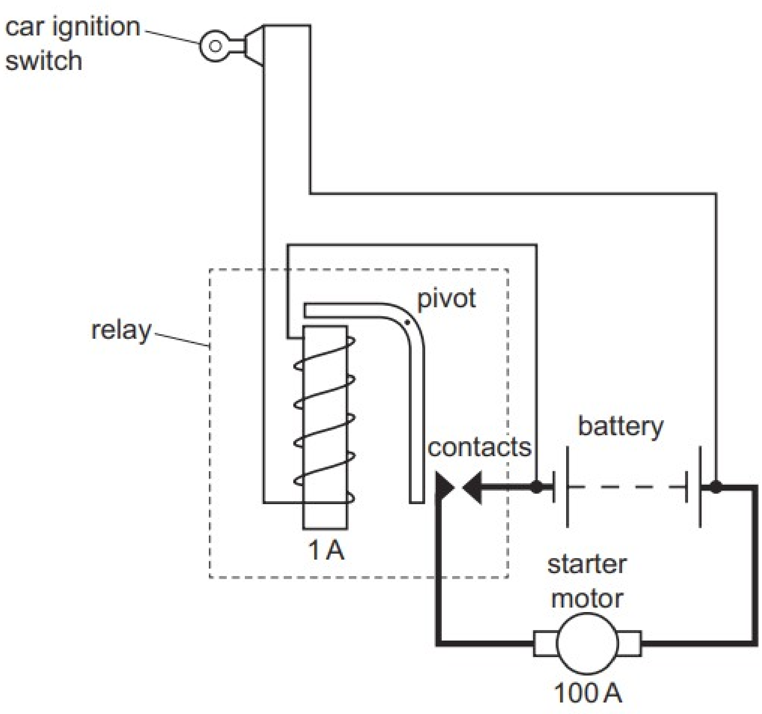
B to decrease the current and increase the potential difference from the power station

C to increase the current and the potential difference from the power station

D to increase the current and decrease the potential difference from the power station

42. M/J/19/12/Q32

The diagram shows a car ignition switch and starter motor.



The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

A Each circuit needs to contain the same total mass of wire.

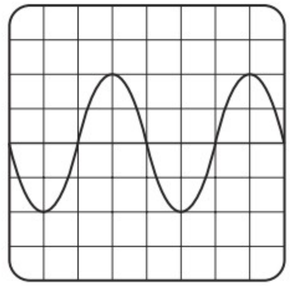
B Thicker wires heat up more quickly when the relay is switched on.

C Thin wires have lower resistances.

D The ignition switch circuit carries a smaller current than the starter motor circuit.

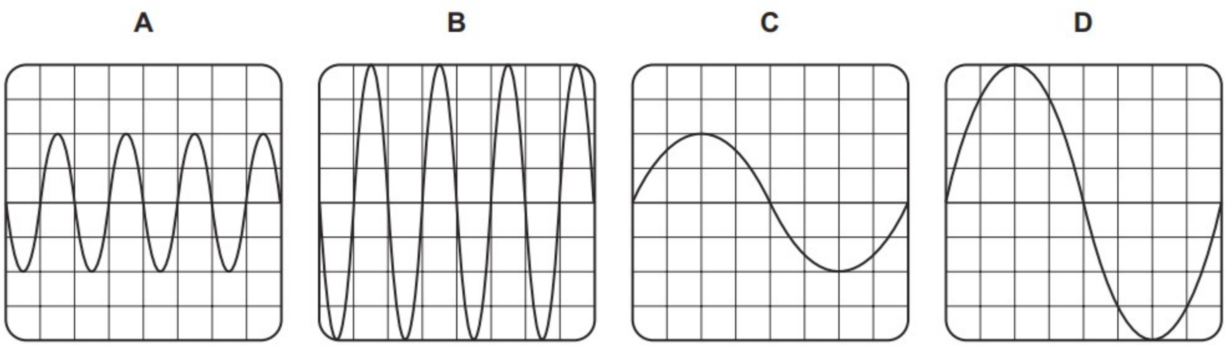
43. M/J/19/12/Q37

A coil is rotating in a magnetic field. The coil is connected to an oscilloscope. The diagram shows the trace on the screen of the oscilloscope.



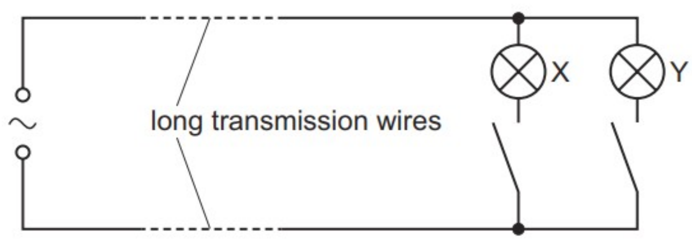
The coil now speeds up and rotates twice as fast.

Which diagram shows the new trace?



44. M/J/19/12/Q38

A teacher uses the circuit shown.



The identical lamps X and Y are connected to a low voltage a.c. power supply by high resistance transmission wires. Both lamps are switched on.

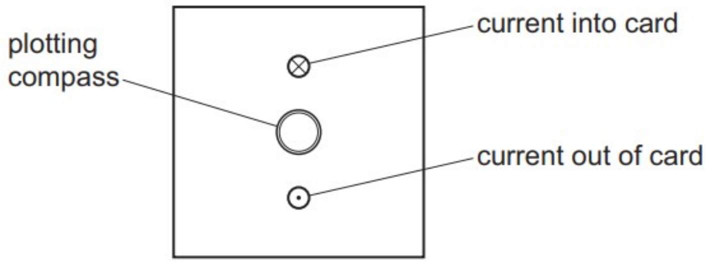
Lamp X is then switched off. Lamp Y stays switched on.

What happens to the voltage and the power supplied to lamp Y?

	the voltage supplied to lamp Y	the power supplied to lamp Y
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

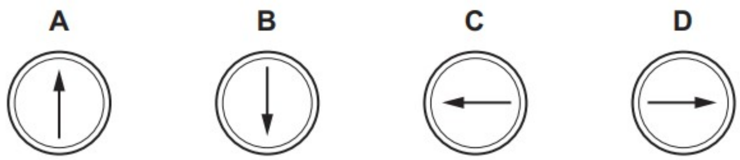
45. M/J/19/11/Q29

Two vertical wires pass at right-angles through a piece of card. There is a large current in each wire in the direction shown.



A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



46. M/J/19/11/Q36

The coil of a simple motor lies between the poles of a permanent magnet. The coil rotates about its axis when there is a current in it.

What decreases the frequency of rotation of the coil?

A increasing the number of turns in the coil

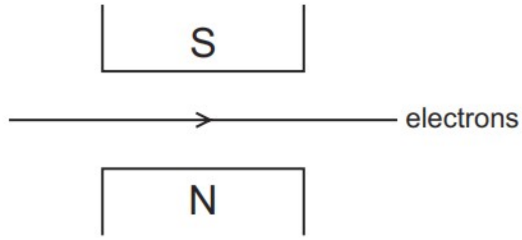
B reversing the current

C using a lower voltage supply

D using a stronger magnet

47. M/J/19/11/Q37

A beam of electrons travels through a vacuum. The beam passes between the poles of a magnet as shown.

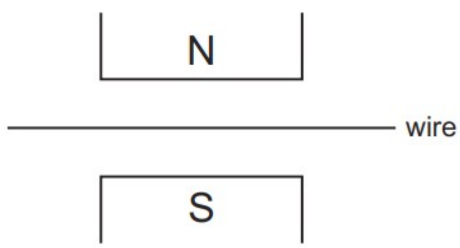


What is the direction of the conventional current and what is the direction of the magnetic field?

	direction of the conventional current	direction of the magnetic field
A	→	↓
B	→	↑
C	←	↓
D	←	↑

48. O/N/18/11/Q33

A wire is placed in the magnetic field between the poles of a magnet, as shown.



When there is a current in the wire, it experiences a force.

Which change increases the size of this force?

A reverse both the direction of the current and the poles of the magnet

B reverse just the direction of the current

C use a stronger magnet

D use a thicker wire with the same current

49. O/N/18/11/Q34

A 12V lamp glows at normal brightness when connected to the secondary coil of a mains transformer. The mains voltage is 240V.

The transformer is 100% efficient and the primary coil has 200 turns.

How many turns are there on the secondary coil?

A 10 **B** 14 **C** 4000 **D** 580 000

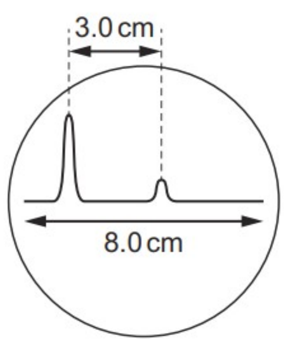
50. O/N/18/11/Q35

Which row shows how the electrical energy produced by a power station is transmitted to distant towns?

	current	voltage
A	alternating	low
B	alternating	very high
C	direct	low
D	direct	very high

51. O/N/18/11/Q36

The diagram shows the trace on an oscilloscope screen. The first peak occurs when a pulse of sound is emitted. The second, smaller peak is the echo received from a wall.



The trace moves across the screen in a time of 24 ms.

How long does it take for the sound to travel to the wall?

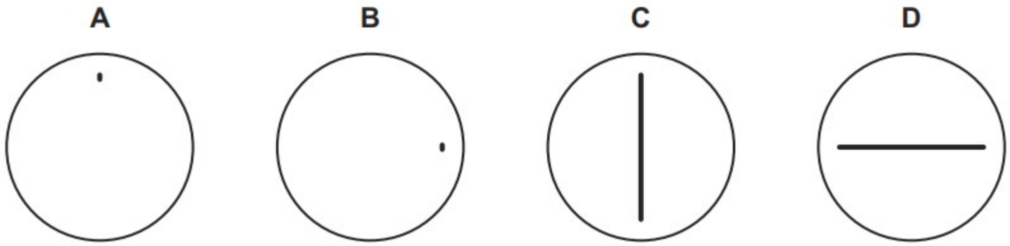
A 4.5 ms **B** 9.0 ms **C** 12 ms **D** 32 ms

52. O/N/18/12/Q36

An oscilloscope is connected to a d.c. power supply.

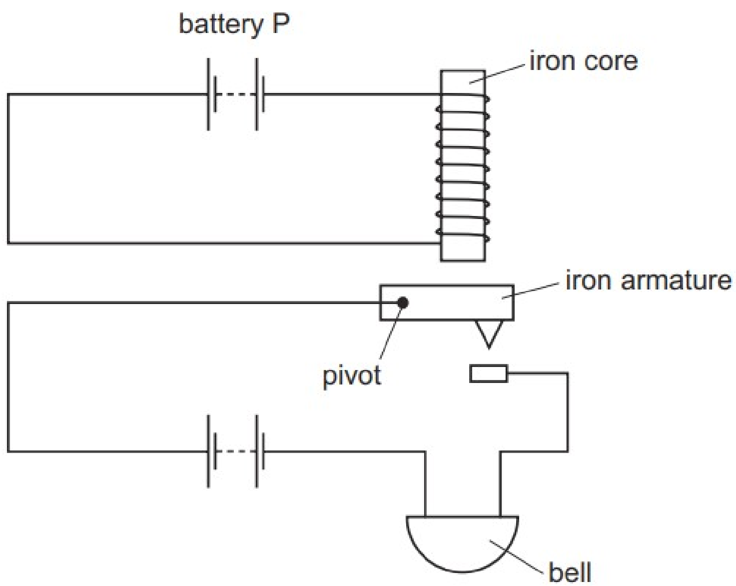
The time-base on the oscilloscope is switched off.

What is seen on the screen of the oscilloscope?



53. M/J/18/12/Q34

The diagram shows an alarm system.



What happens when battery P is disconnected?

	iron armature	bell
A	falls	rings
B	falls	stops ringing
C	moves up	rings
D	moves up	stops ringing

54. M/J/18/12/Q36

A step-down transformer has a primary coil and a secondary coil wound on a soft-iron core.

The primary coil is connected to a 6.0 V direct current (d.c.) supply.

Which statement about the transformer is correct?

A The output voltage is equal to 6.0 V.

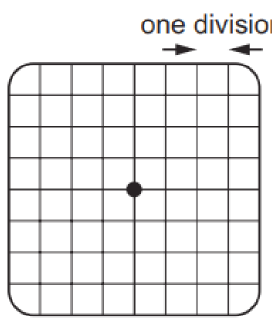
B The output voltage is greater than 6.0 V.

C The output voltage is less than 6.0 V but more than zero.

D There is no output voltage.

55. M/J/18/12/Q39

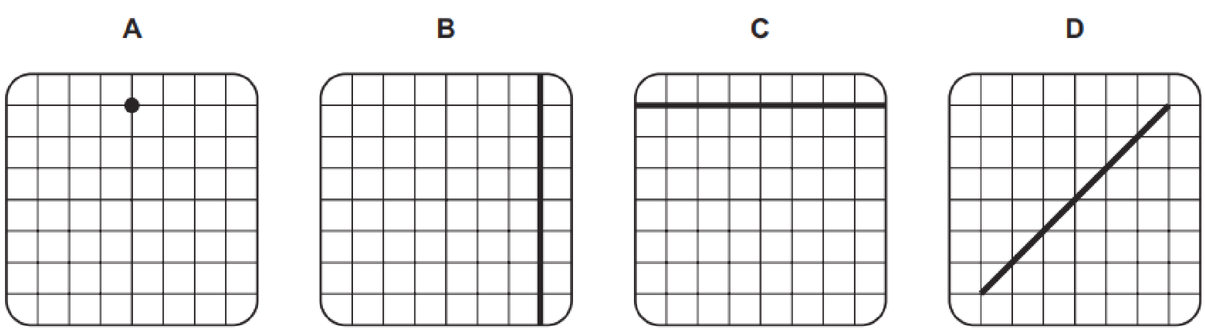
An oscilloscope is used to measure potential difference (p.d.). The trace with no input connected is shown.



A 1.5 V d.c. supply is connected to the oscilloscope.

The Y-gain is set at 0.5 V / div. The time-base is set at 0.5 ms / div.

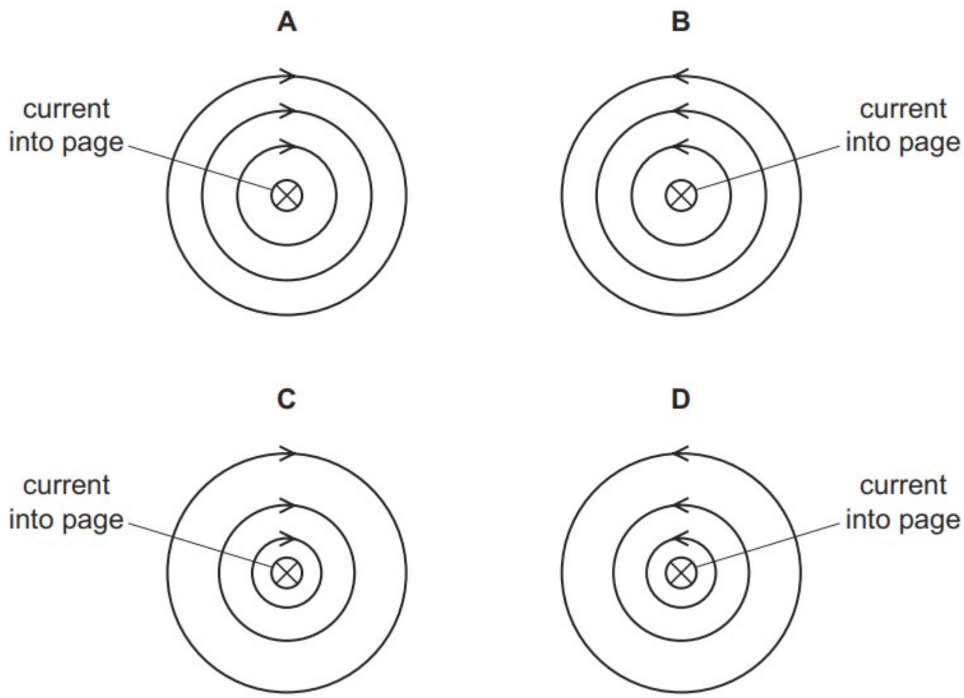
Which trace shows a supply of 1.5 V d.c.?



56. M/J/18/11/Q28

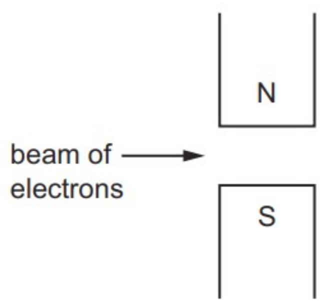
An electric current in a wire is into the page.

Which diagram shows the shape and direction of the magnetic field around the wire?



57. M/J/18/11/Q32

A horizontal beam of electrons passes between the two poles of a magnet.



In which direction is the beam deflected?

A into the page

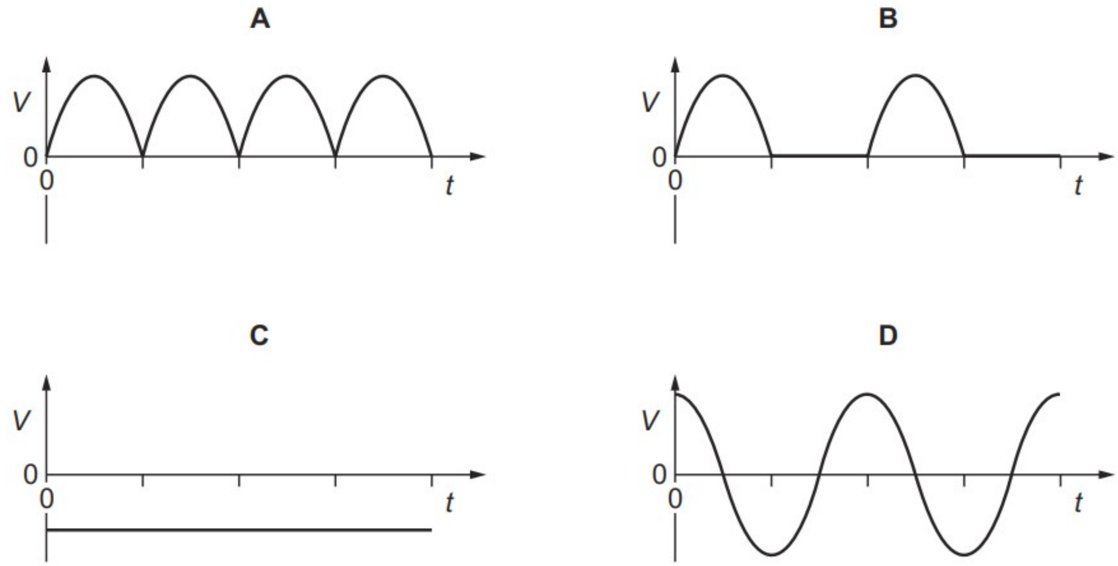
B out of the page

C towards the north pole

D towards the south pole

58. M/J/18/11/Q33

Which graph shows the voltage output V against time t for an a.c. generator?

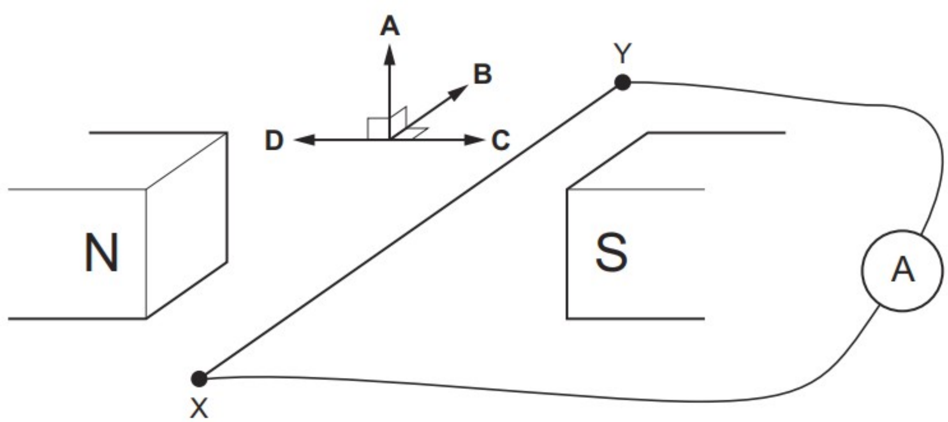


59. O/N/17/11/Q34

The diagram shows a wire XY lying between the poles of a magnet.

The ends of the wire are connected to a sensitive ammeter. The wire is moved and a reading is registered.

In which direction is the wire moved?



60. O/N/17/12/Q33

Why is the coil of an electric motor wound on a soft-iron cylinder?

A to decrease the electric current

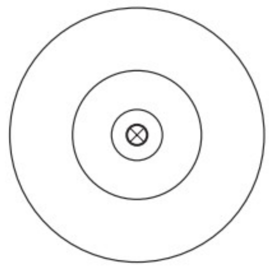
B to increase the electric current

C to reverse the magnetic field

D to strengthen the magnetic field

61. O/N/17/12/Q34

A straight wire carries a current into the paper. The diagram shows three magnetic field lines around the wire.



key
⊗ current into paper

The current in the wire increases.

What is the direction of the field lines and which change occurs to the field lines as the current increases?

	direction	change
A	clockwise	the lines move closer together
B	clockwise	the lines move further apart
C	anticlockwise	the lines move closer together
D	anticlockwise	the lines move further apart

62. M/J/17/12/Q36

What is the effect of using a split-ring commutator?

A it ensures that the current is the same in all parts of a series circuit

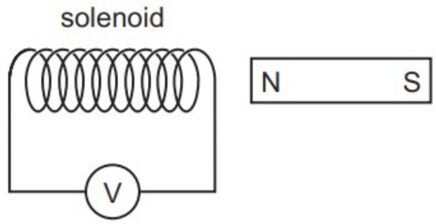
B it generates an alternating electric current

C it produces a force on a current-carrying coil

D it reverses the direction of the current in the coil of a motor

63. M/J/17/12/Q37

The N-pole of a magnet is moved into a solenoid and an e.m.f. is induced.



What causes an increase in the induced e.m.f.?

A moving the magnet more quickly

B moving the magnet more slowly

C pulling the magnet out instead of pushing it in

D using the S-pole of the magnet instead of the N-pole

64. M/J/17/12/Q38

Which material is used for the core of a transformer?

A copper

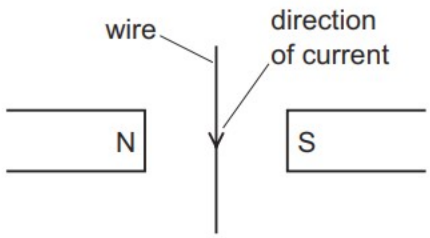
B iron

C plastic

D steel

65. O/N/16/11/Q32

A current-carrying wire lies between the poles of two magnets, as shown.



What is the direction of the force on the wire?

A into the plane of the paper

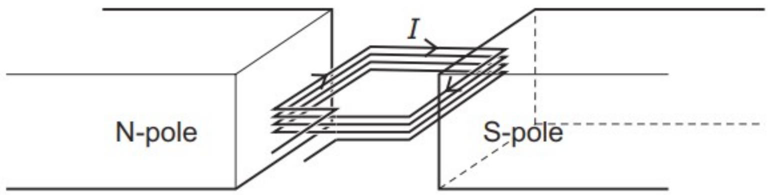
B out of the plane of the paper

C towards the left

D towards the right

66. O/N/16/11/Q33

A coil P of N turns is made from a length L of wire. The coil carries a current I when between two magnetic poles.



A similar coil Q of $2N$ turns is made from a length $2L$ of identical wire. It also carries a current I when between the two magnetic poles.

Which coil has the greater resistance and which coil experiences the greater turning effect?

	greater resistance	greater turning effect
A	P	P
B	P	Q
C	Q	P
D	Q	Q

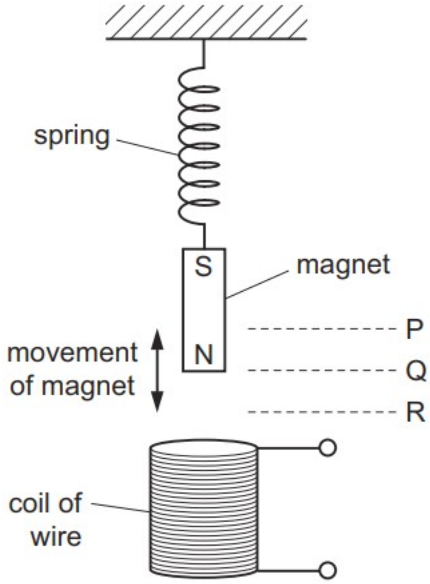
67. O/N/16/11/Q35

Which material is most suitable for the core of a transformer and which material is most suitable for the coils in the transformer?

	material for core	material for coils
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

68. O/N/16/11/Q34

A magnet moves up and down above a coil of wire.



The bottom of the magnet moves up and down between P and R.

Where is the bottom of the magnet when there is no induced electromotive force (e.m.f.) in the coil?

A at P and at Q

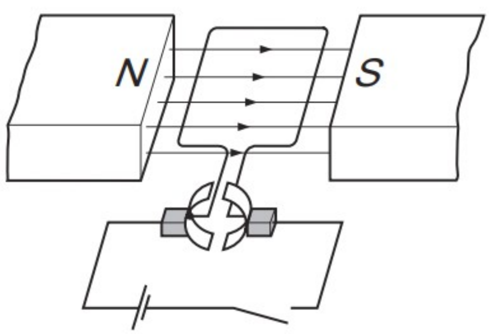
B at P and at R

C at Q only

D at R only

69. M/J/16/12/Q32

The diagram shows a simple d.c. motor.



The switch is closed and the coil rotates.

Which change makes the coil rotate in the opposite direction **and** at a faster rate?

A increase the current in the coil and increase the number of turns in the coil

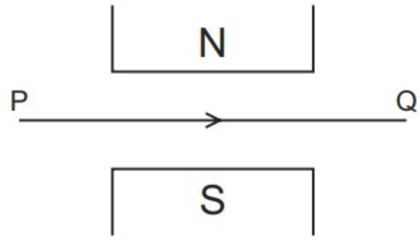
B reverse both the magnetic field and the current in the coil

C reverse the magnetic field and decrease the current in the coil

D reverse the magnetic field and increase the current in the coil

70. M/J/16/12/Q33

The diagram shows a wire PQ between the N-pole and the S-pole of a magnet. There is a current in the wire in the direction of the arrow.



What is the direction of the force on the wire PQ?

A into the page

B out of the page

C towards the N-pole

D towards the S-pole

71. M/J/16/12/Q34

Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

72. M/J/16/11/Q35

A split-ring commutator is used in a simple d.c. motor. It reverses the current in the coil.

How often does it reverse the current?

A every quarter turn

B every half turn

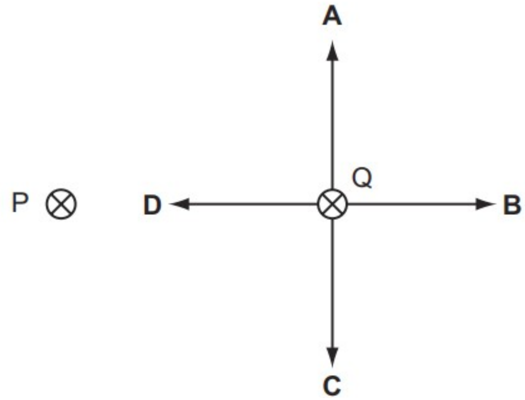
C every full turn

D every two turns

73. O/N/15/11/Q32

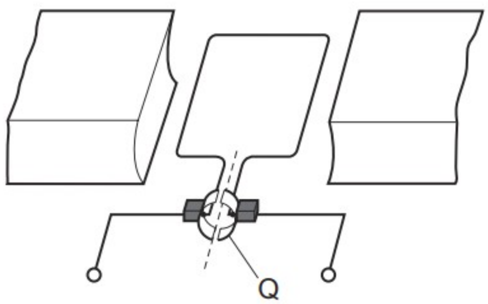
P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



74. O/N/15/11/Q33

The diagram shows a simple d.c. motor.



What is the part labelled Q?

A a coil

B a commutator

C a magnet

D a slip ring

75. O/N/15/11/Q35

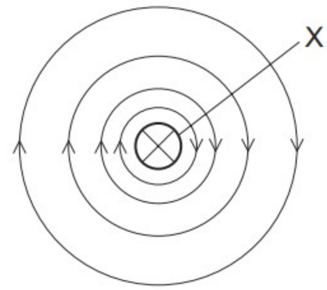
Electric power cables transmit electrical energy over large distances using high-voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	high current is produced in the cable	the resistance of the cable is reduced
B	high current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

76. O/N/15/12/Q31

The diagram shows the magnetic field around wire X which carries a current into the paper.



The arrows on the field lines show the direction of the force on

A a N-pole.

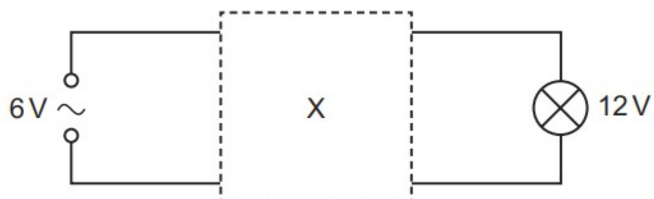
B a S-pole.

C a small negative charge.

D a small positive charge.

77. O/N/15/12/Q37

The diagram shows an electrical device X connected between a 6 V a.c. supply and a 12 V lamp



The lamp is seen to glow with normal brightness.

What is X?

A a capacitor

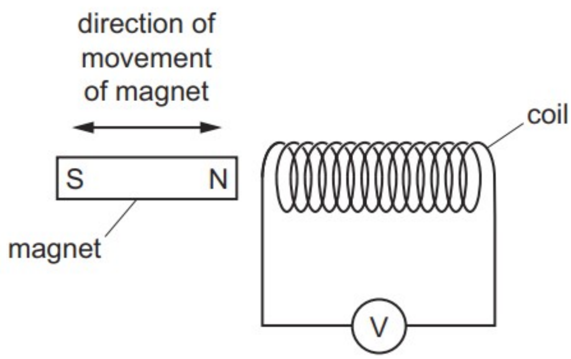
B a potential divider

C a relay

D a transformer

78. M/J/15/12/Q36

A teacher moves a magnet into and out of a coil of wire, as shown, in order to demonstrate electromagnetic induction.



Which statement is correct?

A As the magnet is moved into the coil the left-hand end of the coil becomes a S-pole.

B As the magnet is taken out of the coil the left-hand end of the coil becomes a N-pole.

C Increasing the speed at which the magnet enters the coil, increases the induced voltage.

D Increasing the speed at which the magnet leaves the coil decreases the induced voltage.

79. M/J/15/12/Q37

A transformer consists of two coils which are wound on to a metallic core.

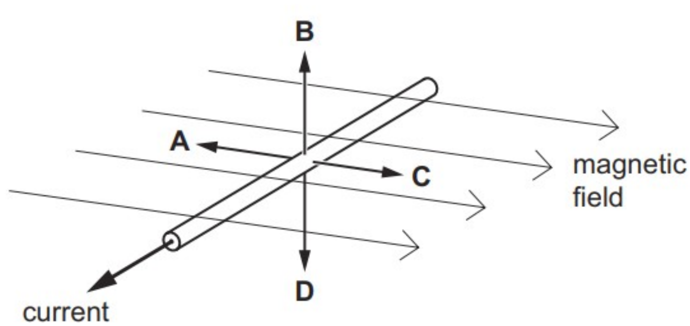
Which type of voltage is supplied to the transformer and which metal is used to make the core?

	supply voltage	metal
A	alternating	iron
B	alternating	steel
C	direct	iron
D	direct	steel

80. O/N/14/11/Q30

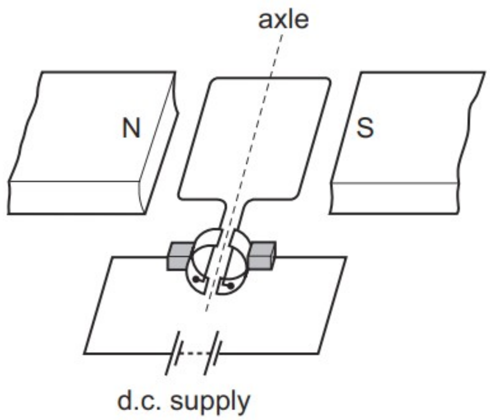
The diagram shows a current-carrying wire in a horizontal magnetic field.

Which arrow shows the direction of the force experienced by the wire?



81. O/N/14/11/Q31

The diagram shows a d.c. motor with its coil horizontal.



Why is a split-ring commutator used?

A to change the current direction in the coil as the coil passes the horizontal position

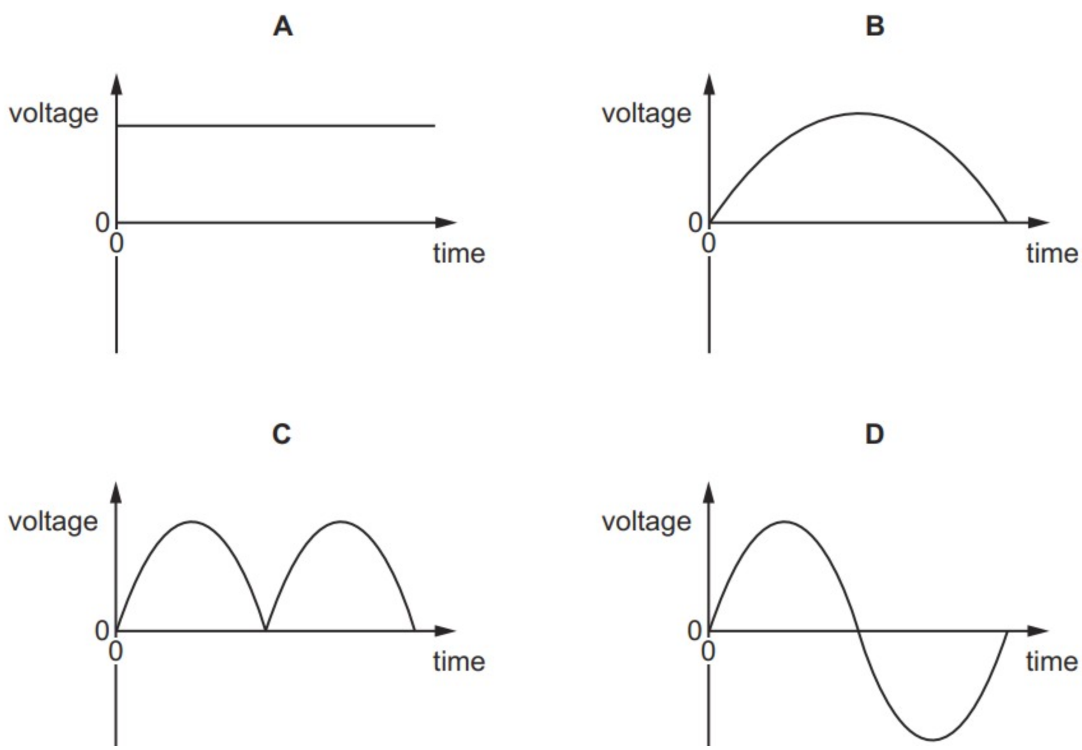
B to change the current direction in the coil as the coil passes the vertical position

C to change the current direction in the d.c. supply as the coil passes the horizontal position

D to change the current direction in the d.c. supply as the coil passes the vertical position

82. O/N/14/11/Q32

Which graph shows the voltage output of an a.c. generator when the coil makes **one** complete revolution?



83. O/N/14/11/Q36

Why is a reed relay used in a switching circuit?

A to switch on a small current using a large current

B to switch on a small voltage using a large voltage

C to switch on a large current using a small current

D to switch on a large voltage using a large current

84. O/N/14/12/Q31

Two long, parallel conductors carrying current lie in a horizontal plane.

The two conductors attract one another.

The two currents **must**

A be in the same direction.

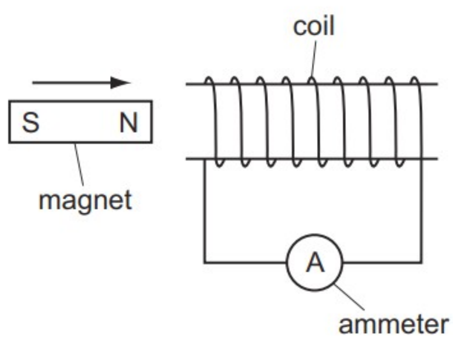
B be in opposite directions.

C be parallel to the Earth's magnetic field.

D be at 90° to the Earth's magnetic field.

85. O/N/14/12/Q33

A student moves a magnet into a coil of wire as shown in the diagram. The coil of wire is connected to a sensitive ammeter.



Which change does **not** produce an increase in the reading?

A increasing the number of turns on the coil

B increasing the resistance of the ammeter

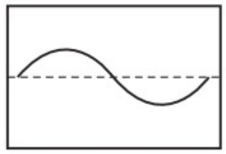
C increasing the speed of the magnet

D increasing the strength of the magnet

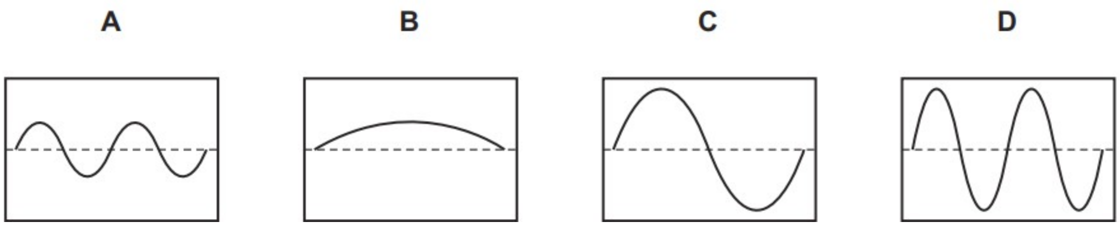
86. O/N/14/12/Q34

The coil of an a.c. generator is rotated and the output is displayed on the screen of a cathode-ray oscilloscope (c.r.o.).

The diagram shows the trace on the screen.



Which trace appears on the screen when the speed of rotation of the coil is doubled but the settings on the c.r.o. are unaltered?



87. M/J/14/12/Q34

Which device uses the force experienced by a current in a magnetic field when in normal use?

A cathode-ray oscilloscope

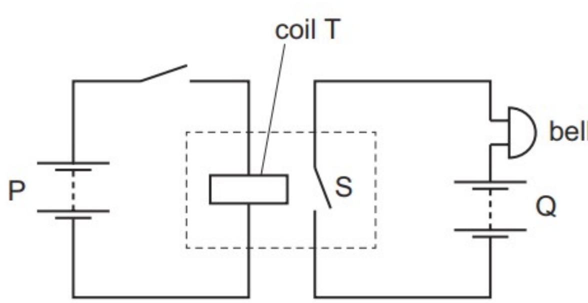
B electrostatic precipitator

C loudspeaker

D transformer

88. M/J/14/12/Q35

A relay is used in a circuit containing a bell.



How can the apparatus be altered to make the sound of the bell louder?

A increase the number of turns on coil T

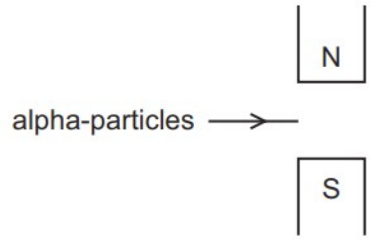
B increase the voltage of battery P

C increase the voltage of battery Q

D move the coil closer to switch S

89. M/J/14/11/Q34

A beam of alpha-particles enters the magnetic field between the poles of a magnet.



In which direction is the magnetic force on the beam?

A down the page

B into the page

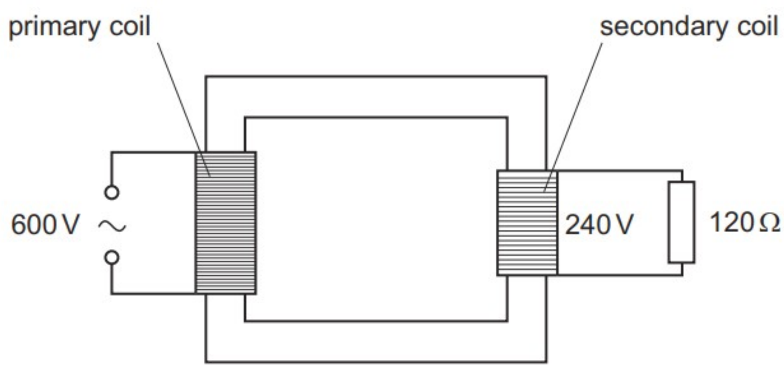
C out of the page

D up the page

90. M/J/14/11/Q36

An ideal transformer has a primary voltage of 600 V and a secondary voltage of 240 V.

The secondary coil is attached to a resistor of resistance 120 Ω.

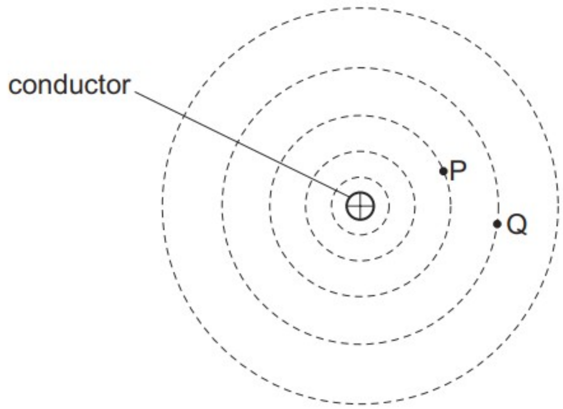


What is the power dissipated in the resistor and the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

91. O/N/13/11/Q25

The diagram shows the shape of the magnetic field lines near a current-carrying conductor.



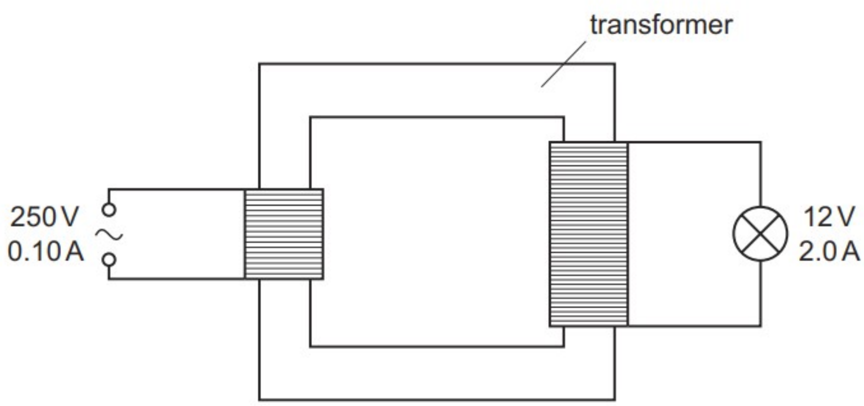
The current in the conductor is into the plane of the diagram.

Which row correctly states the direction of the field lines and compares the strengths of the field at points P and Q?

	direction of field lines	the field is stronger at
A	clockwise	P
B	clockwise	Q
C	anticlockwise	P
D	anticlockwise	Q

92. O/N/13/11/Q31

A transformer is used to operate a 12 V lamp from a 250 V mains supply.



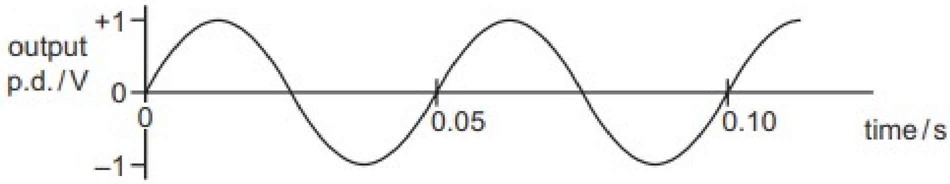
The mains current is 0.10 A. The current in the lamp is 2.0 A.

What is the efficiency of the transformer?

A 0.048 **B** 0.050 **C** 0.96 **D** 1.04

93. O/N/13/11/Q33

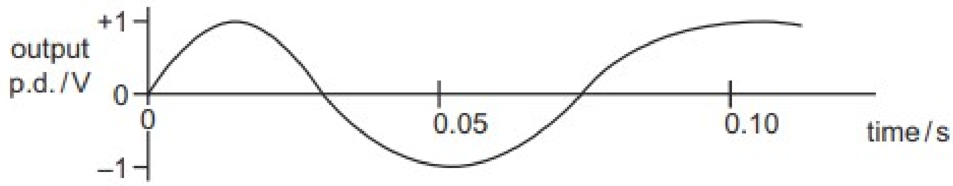
The graph shows the output of an a.c. generator. The coil in the generator rotates 20 times in one second.



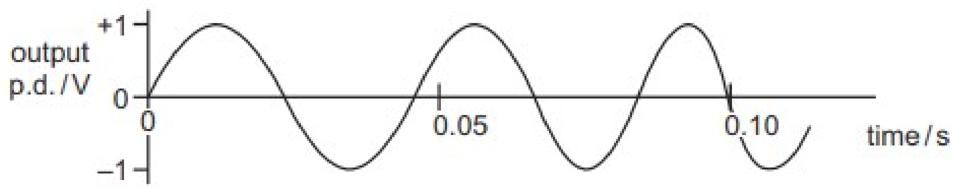
The speed of rotation of the coil steadily increases.

Which graph best shows how the output changes?

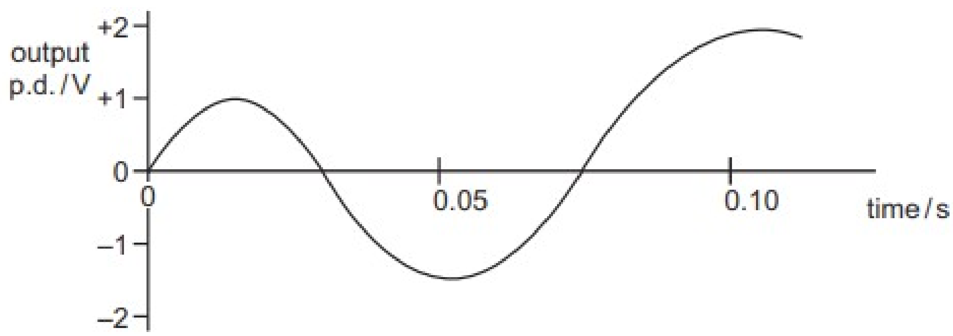
A



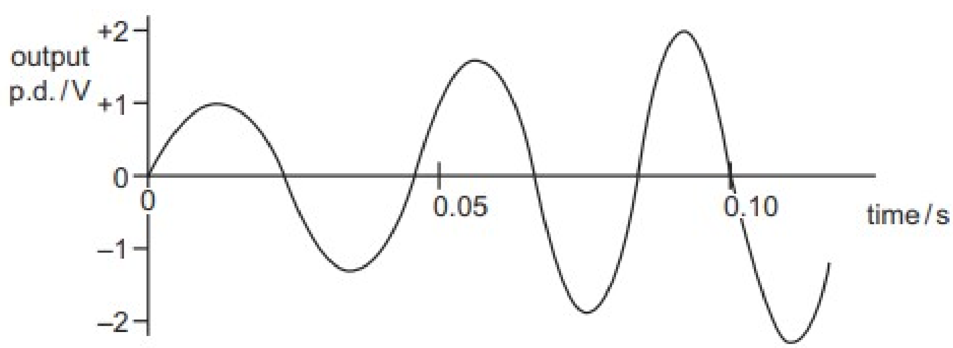
B



C



D



94. O/N/13/11/Q34

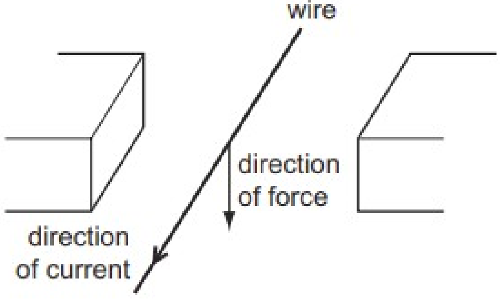
Which material is used for the core of a transformer and why?

	material	reason
A	copper	good conductor of electricity
B	copper	easy to magnetise and demagnetise
C	iron	good conductor of electricity
D	iron	easy to magnetise and demagnetise

95. O/N/13/12/Q32

The diagram shows a wire placed between two magnetic poles of equal strength.

A current passes through the wire in the direction shown. The current causes a downward force on the wire.



What is the arrangement of the magnetic poles?

A

B

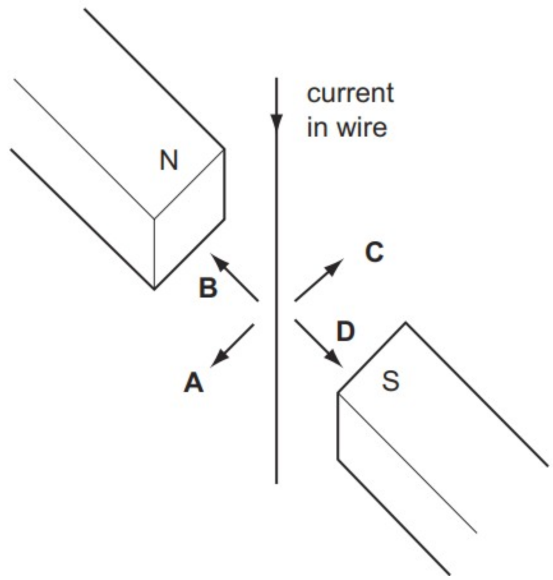
C

D

96. M/J/13/12/Q37

A wire hangs between the poles of a magnet.

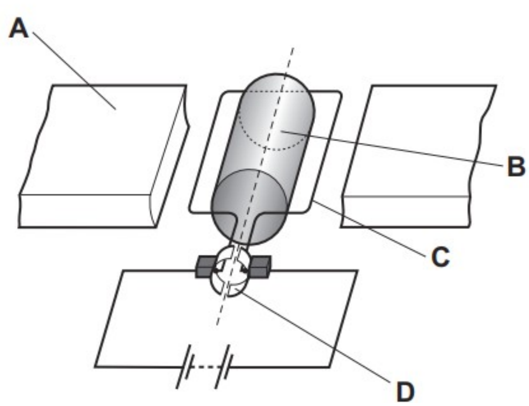
When there is a current in the wire, in which direction does the wire move?



97. M/J/13/11/Q35

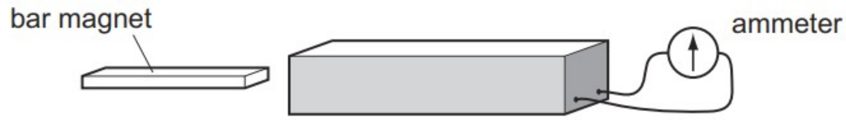
The diagram shows a simple d.c. motor.

Which labelled part is the commutator?



98. M/J/13/11/Q36

Two leads emerging from a box are connected to a sensitive ammeter.



When a bar magnet moves towards the open end of the box, the needle of the ammeter deflects to the right. When the bar magnet stops, the needle returns to zero.

What is inside the box?

A a coil alone

B a coil connected in series with a cell

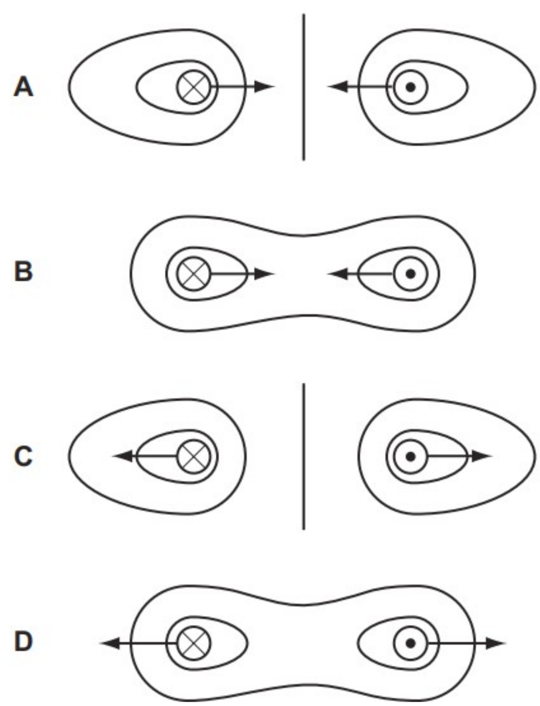
C a light-dependent resistor (LDR) alone

D an LDR in series with a cell

99. O/N/12/11/Q33

Each of the diagrams shows a cross-section through two parallel, current-carrying conductors.

Which diagram shows the shape of the magnetic field pattern **and** the directions of the forces on the two conductors?



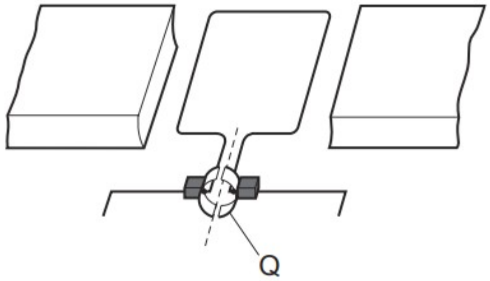
key

⊗ conductor carrying current into page

⊙ conductor carrying current out of page

100. O/N/12/22/Q34

The diagram shows a simple d.c. motor.



What is the part labelled Q?

A a coil

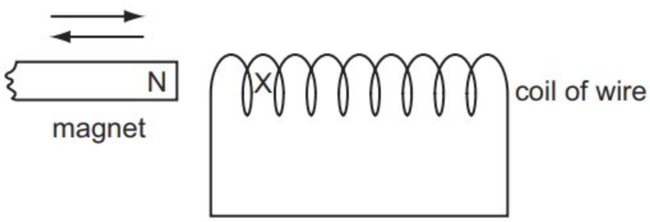
B a magnet

C a slip ring

D a split-ring commutator

101. O/N/12/11/Q35

The diagram shows the N-pole of a magnet moving into, and out of, a coil of wire.



This movement produces a current in the coil of wire. The current produces a magnetic pole at X.

Which pole is produced at X when the magnet is moved in and when it is moved out?

	magnet moved in	magnet moved out
A	N	N
B	N	S
C	S	N
D	S	S

102. O/N/12/11/Q37

What is required to operate a reed relay in a switching circuit?

A a capacitor

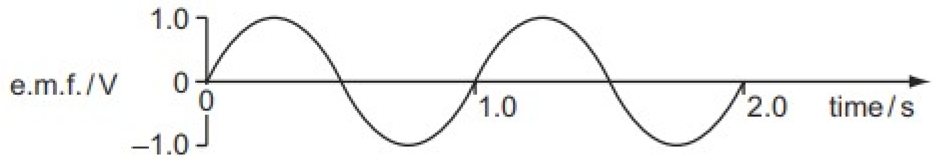
B an electric field

C a magnetic field

D a transformer

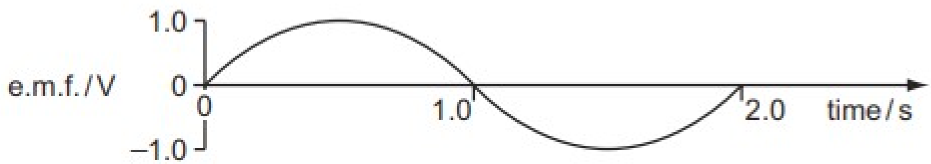
103. O/N/12/11/Q36

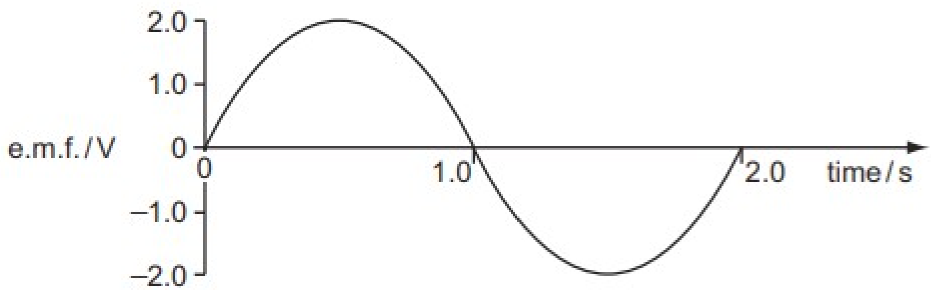
A simple a.c. generator produces an alternating e.m.f. as shown.

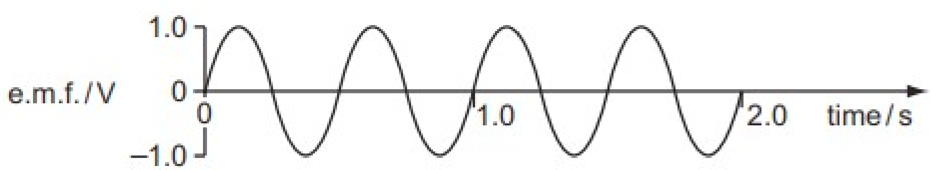


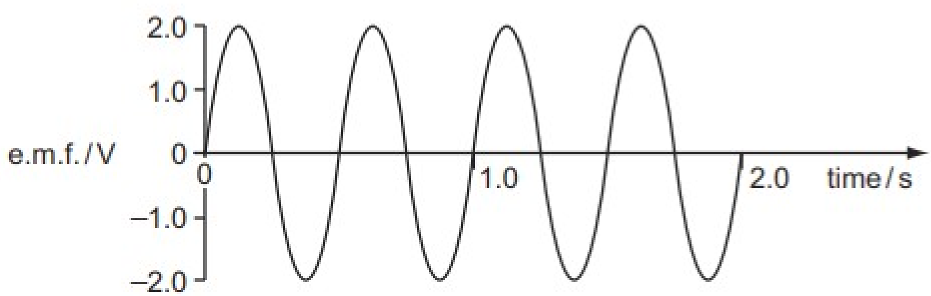
The speed of the generator is doubled.

Which graph best represents the new output?

A


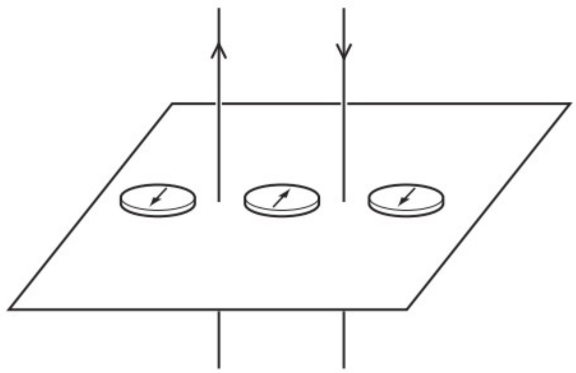
B


C


D


104. O/N/12/12/Q33

Two parallel wires carry currents in opposite directions. Three plotting compasses are placed in the positions shown.



The currents in **both** wires are reversed. How many compass needles change direction? (Ignore the effect of the Earth's magnetic field.)

A 0 **B** 1 **C** 2 **D** 3

105. O/N/12/12/Q34

Which single-coil motor has the largest turning effect?

	current in coil / A	number of turns in coil	iron core
A	6	100	no
B	10	200	no
C	6	100	yes
D	10	200	yes

106. O/N/12/12/Q35

A magnet is moved towards a coil of insulated wire. A voltmeter connected across the coil shows a positive reading.

What produces a higher reading on the voltmeter?

A moving the magnet away from the coil at the same speed

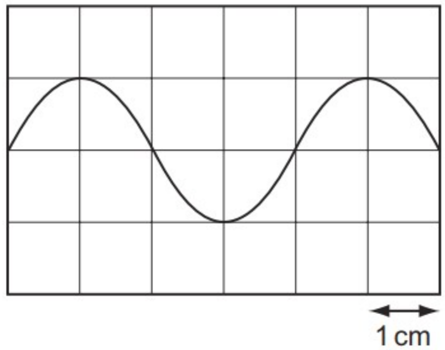
B moving the magnet away from the coil at a slower speed

C moving the magnet towards the coil at a faster speed

D moving the magnet towards the coil at a slower speed

107. O/N/12/12/Q36

The diagram shows the output of an a.c. generator as displayed on a cathode-ray oscilloscope. The horizontal scale is 5 ms / cm.

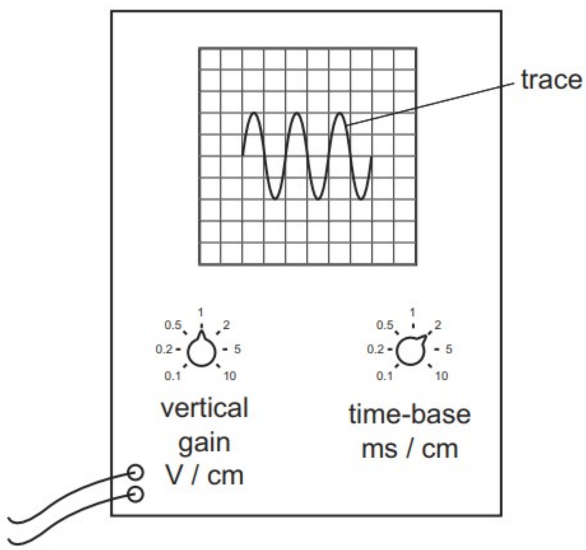


What is the time for one complete rotation of the coil of the generator?

A 5 ms **B** 10 ms **C** 20 ms **D** 30 ms

108. O/N/12/12/Q37

The trace of a waveform is seen on the screen of a cathode-ray oscilloscope.



Which statement about the controls is correct?

A The amplitude of the trace is changed by adjusting the time-base.

B The amplitude of the trace is changed by adjusting the vertical gain.

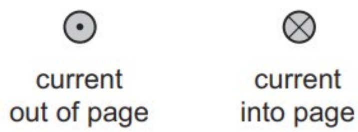
C The whole trace is moved to the right by adjusting the time-base.

D The whole trace is moved upwards by adjusting the vertical gain.

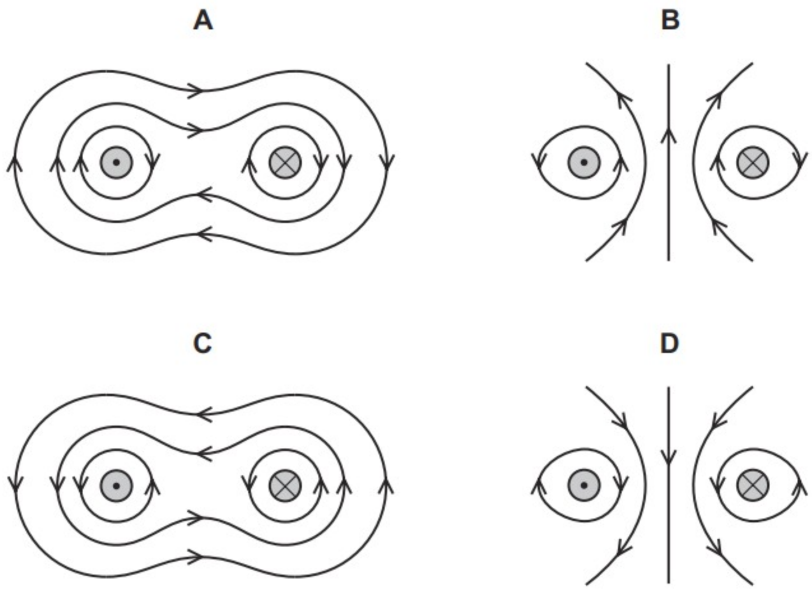
109. M/J/12/12/Q26

Two long, straight wires hang vertically, close to each other.

The wires carry currents in opposite directions.

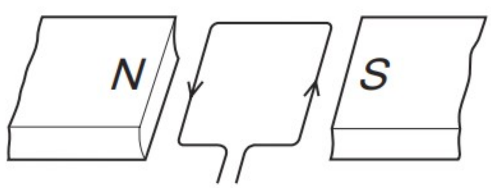


Which diagram shows the magnetic field pattern around the wires?



110. M/J/12/12/Q34

A rectangular coil is placed between the poles of a magnet. A current passes through the coil, as shown.



What happens to the coil?

A It moves downwards.

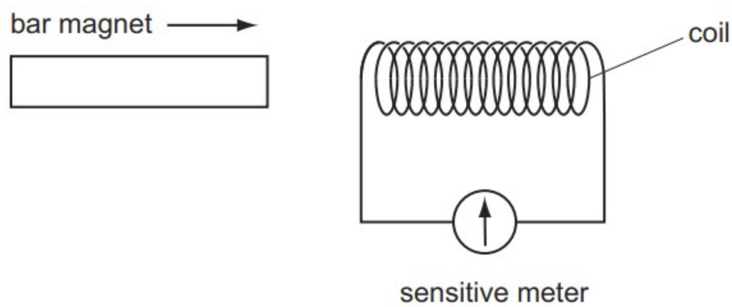
B It moves upwards.

C It rotates anticlockwise.

D It rotates clockwise.

111. M/J/12/12/Q35

A bar magnet is pushed into one end of a long coil connected to a sensitive meter.



Which of the following affects the **magnitude** of the deflection of the meter?

A the direction in which the coil is wound

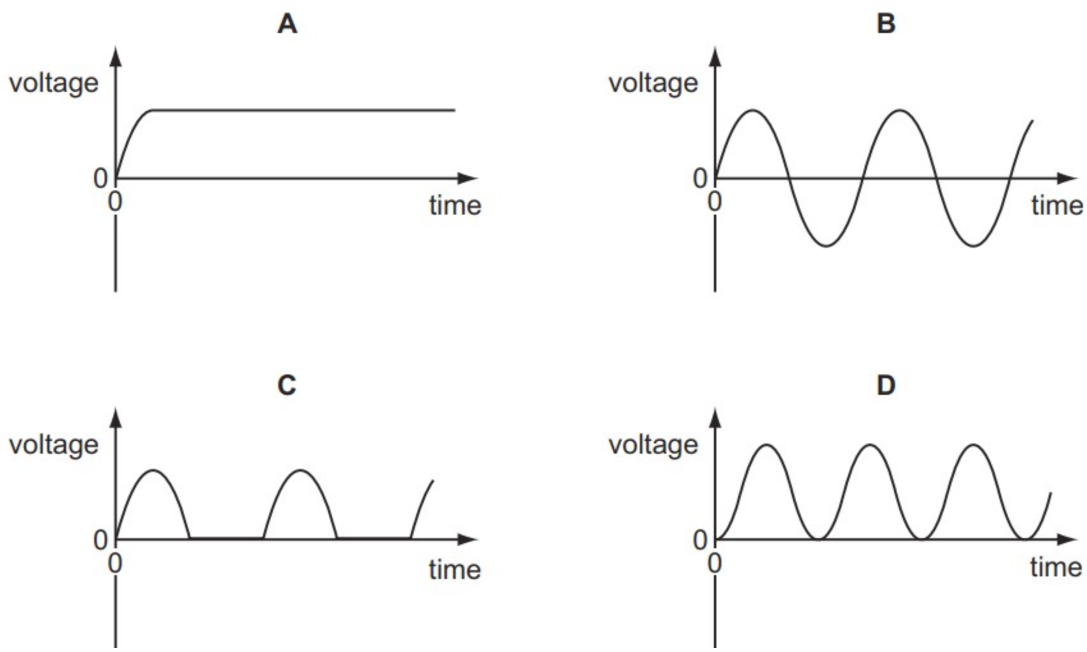
B the speed with which the magnet enters the coil

C which end of the coil is used

D which pole of the magnet enters first

112. M/J/12/12/Q36

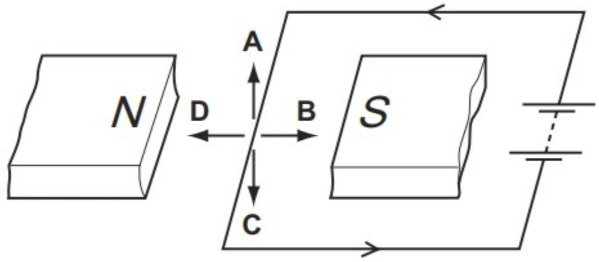
Which graph represents the voltage output of a simple a.c. generator?



113. M/J/12/11/Q33

A current-carrying wire is placed between the poles of a magnet.

What is the direction of the force on the wire due to the current?



114. M/J/12/11/Q34

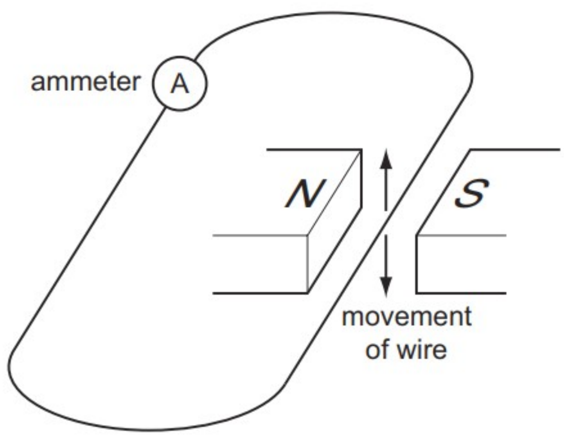
A simple d.c. motor consists of a coil that rotates between the poles of a permanent magnet. The turning effect is increased by winding the coil on a metal cylinder.

Which metals are used to make the magnet and the cylinder?

	magnet	cylinder
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

115. M/J/12/11/Q35

A current is produced when a wire is moved between two magnets as shown.



Which device uses this effect?

A a battery

B a generator

C a motor

D an electromagnet

116. M/J/12/11/Q36

Why is a transformer used to connect a generator in a power station to a long-distance transmission line?

A to decrease the voltage and decrease the current

B to decrease the voltage and increase the current

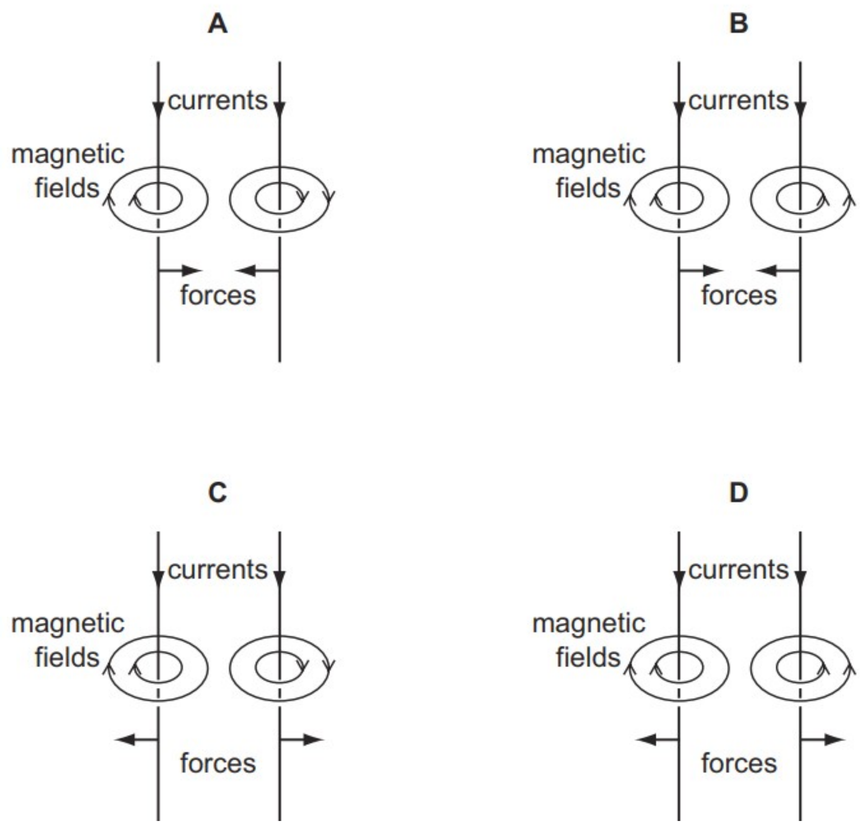
C to increase the voltage and decrease the current

D to increase the voltage and increase the current

117. O/N/11/11/Q34

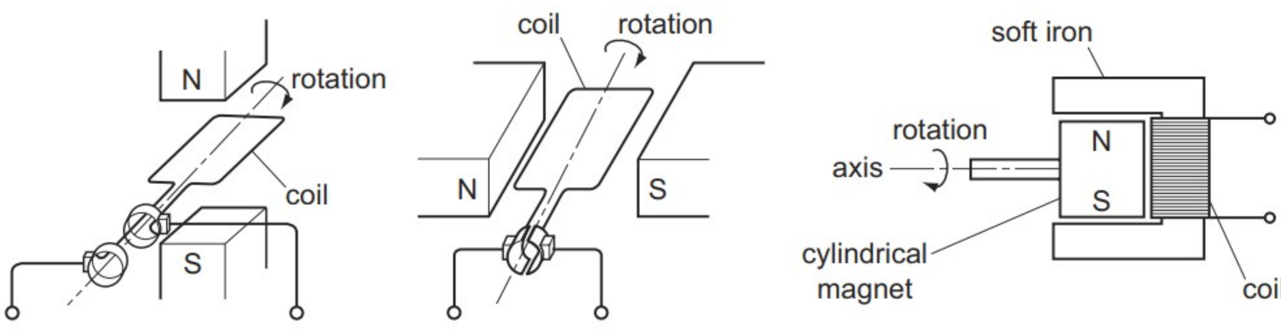
Two parallel wires carry currents in the same direction.

Which diagram shows the magnetic field around each wire and the direction of the force on each wire?



118. O/N/11/11/Q35

The diagrams show three electrical devices, X, Y and Z.



X Y Z

Which devices provide an alternating current (a.c.) output?

A X only **B** Y only **C** X and Y **D** X and Z

119. O/N/11/11/Q36

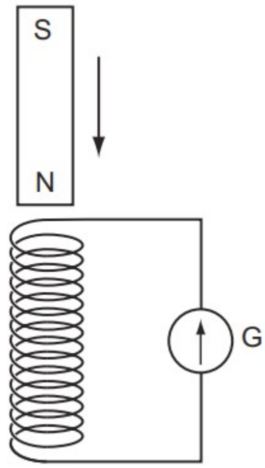
One component of a simple d.c. motor is a split-ring commutator.

Which metal is used to make the commutator, and why is this metal chosen?

	metal	reason
A	copper	it is a good conductor of electricity
B	copper	it is a good conductor of heat
C	iron	it increases the magnetic field strength
D	iron	it is attracted to the brushes

120. O/N/11/11/Q37

A small coil is connected to a galvanometer G, as shown.



When a magnet is allowed to fall towards the coil, the galvanometer pointer gives a momentary deflection to the right of the zero position.

The magnet moves through the coil.

What happens to the galvanometer pointer as the magnet falls away from the coil?

A It gives a continuous reading to the left.

B It gives a momentary deflection to the left.

C It gives a continuous reading to the right.

D It gives a momentary deflection to the right.

121. M/J/11/12/Q34

The electromotive force (e.m.f.) induced in a conductor moving at right-angles to a magnetic field does **not** depend upon

A the length of the conductor.

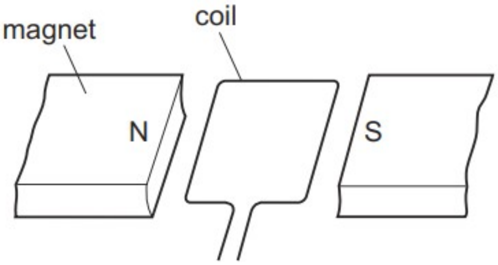
B the resistance of the conductor.

C the speed of the conductor.

D the strength of the magnetic field.

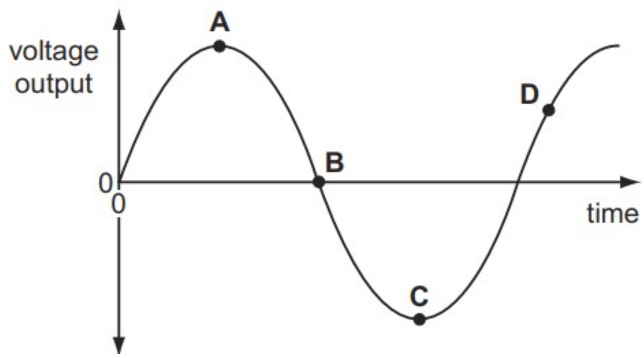
122. M/J/11/12/Q33

The diagram shows part of an a.c. generator when its coil is in a horizontal position.



The graph shows the voltage output plotted against time.

Which point on the graph shows when the coil is in a vertical position?



123. M/J/11/12/Q35

The coil in an electric motor is wound onto a cylinder.

Why is the cylinder made of soft iron?

A to deflect the magnetic field away from the coil

B to increase the current through the coil

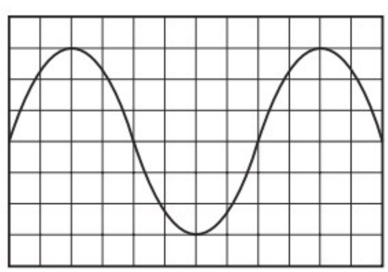
C to increase the strength of the magnetic field through the coil

D to support the coil and prevent it from collapsing

124. M/J/11/12/Q37

An alternating voltage of frequency 0.5 Hz is applied to the Y-plates of a cathode-ray oscilloscope (c.r.o.).

The diagram shows the screen of the c.r.o.

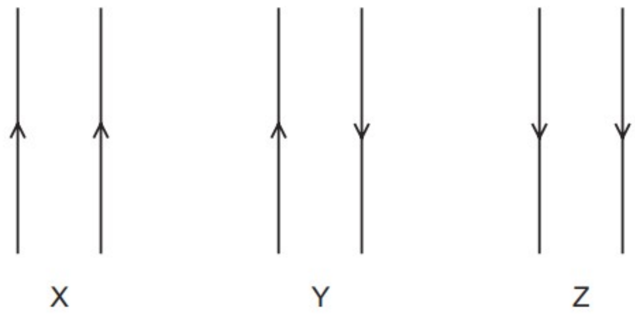


What is the time taken for the spot to cross the screen?

A 3 s **B** 6 s **C** 15 s **D** 30 s

125. M/J/11/12/Q36

The diagram shows three pairs of parallel wires with the currents in the directions shown.



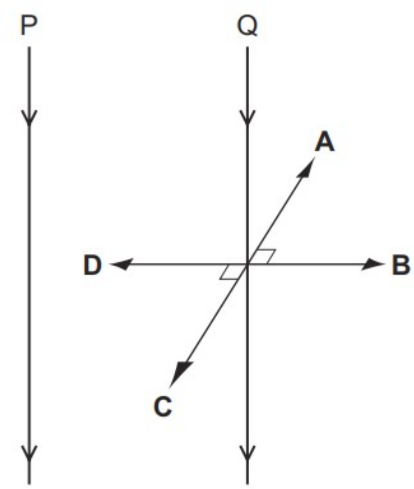
For each pair of wires, what are the forces between the wires?

	X	Y	Z
A	attraction	none	repulsion
B	attraction	repulsion	attraction
C	repulsion	attraction	repulsion
D	repulsion	repulsion	repulsion

126. O/N/10/11/Q32

Two parallel vertical wires P and Q are a small distance apart in air. There is a downwards electric current in both wires. A force acts on Q owing to the current in P. This force is perpendicular to the wire Q.

What is the direction of the force on Q?



127. O/N/10/11/Q33

What does **not** alter the size of the turning effect on the coil of an electric motor?

A the direction of the current in the coil

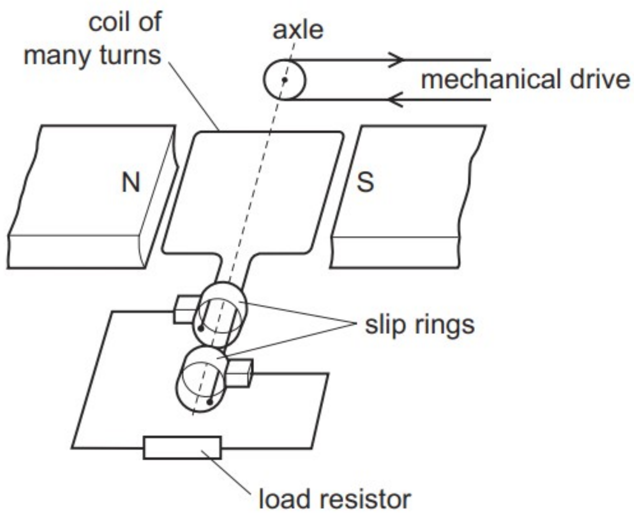
B the number of turns in the coil

C the size of the current in the coil

D the strength of the magnetic field

128. O/N/10/11/Q34

The diagram shows an a.c. generator connected to an electrical circuit (load resistor).



Which statement is correct?

A The direction of the potential difference across the load resistor is always the same.

B The size of the induced e.m.f. depends on the number of turns in the coil.

C The size of the induced e.m.f. does not change as the coil turns.

D Winding the coil on a soft-iron cylinder makes no difference to the induced e.m.f.

129. O/N/10/11/Q35

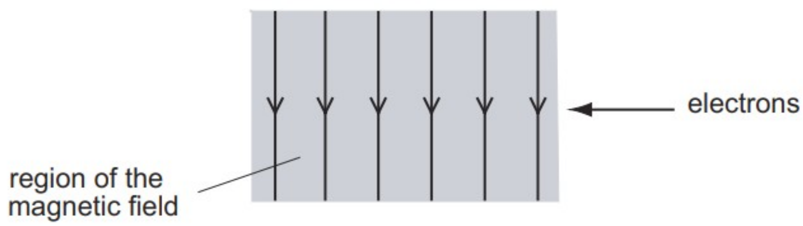
Electric power cables transmit electrical energy over large distances using a high voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	a higher current is produced in the cable	the resistance of the cable is reduced
B	a higher current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

130. O/N/10/11/Q36

The diagram shows a beam of electrons entering a magnetic field. The direction of the magnetic field is downwards, towards the bottom of the page.



In which direction does the deflection of the electrons occur?

A into the page

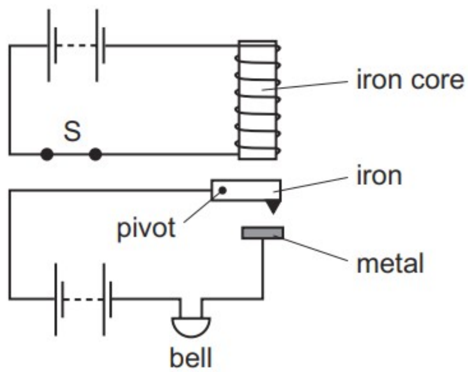
B out of the page

C towards the bottom of the page

D towards the top of the page

131. O/N/10/11/Q37

The diagram shows an alarm system in which the switch S is shown closed.



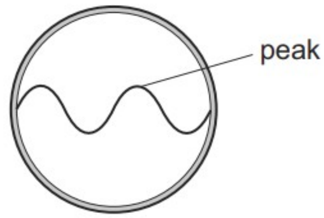
What happens when the switch S is opened?

	iron	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

132. M/J/10/12/Q32

A cathode-ray oscilloscope is connected to an a.c. generator.

A wave is seen on the screen of the oscilloscope.



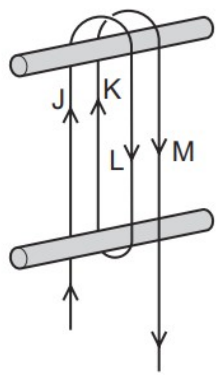
The speed of rotation of the generator is doubled.

What is the effect on the wave?

	number of peaks on the screen	amplitude of wave on the screen
A	doubled	doubled
B	doubled	same
C	same	doubled
D	same	same

133. M/J/10/12/Q33

A long flexible wire is wrapped round two wooden pegs. A large current is passed in the direction shown.



Which two pairs of lengths of wire attract each other?

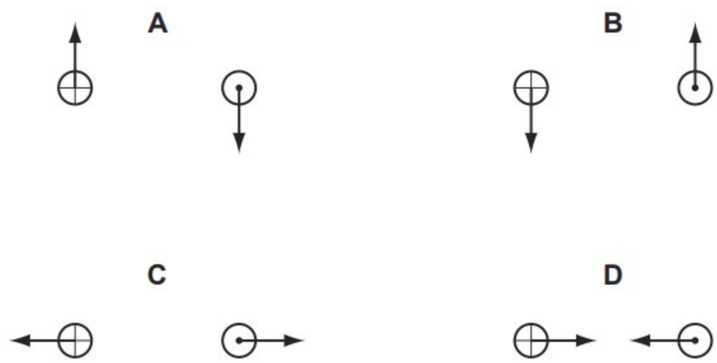
	first pair	second pair
A	J and K	K and M
B	J and K	L and M
C	J and L	K and M
D	J and L	L and M

134. O/N/09/Q31

Each diagram shows a cross-section through two parallel conductors, each carrying an electric current.

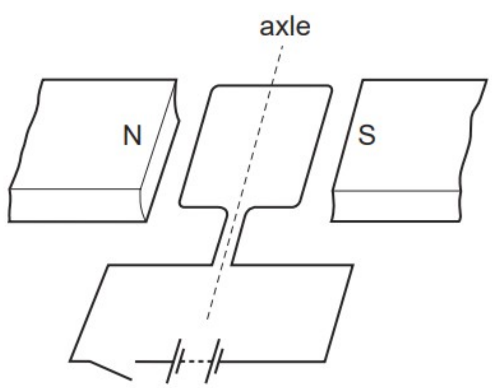
In the conductor on the left, the current is into the page; on the right, it is out of the page.

Which diagram shows the directions of the forces on the two conductors?



135. O/N/09/Q32

A simple model of a d.c. motor is made. By mistake, the split-ring commutator is left out. The coil can turn, but is always connected to the battery in the same way.



The coil starts in the horizontal position.

What happens to the coil when the circuit is switched on?

A It does not move at all.

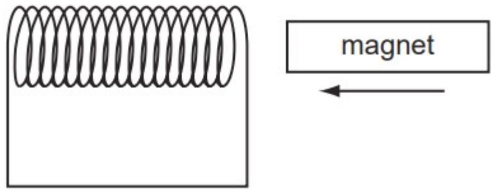
B It moves upwards, out of the magnetic field.

C It turns to the vertical position and eventually stops there.

D It turns to the vertical position then comes back to the horizontal position.

136. O/N/09/Q33

A magnet is pushed horizontally towards a coil of wire, inducing an e.m.f. in the coil.



In which direction does the induced e.m.f. make the coil move?

A away from the magnet

B towards the magnet

C downwards

D upwards

137. O/N/09/Q35

Why is a transformer used to connect a generator in a power station to a long-distance transmission line?

A to decrease the voltage and decrease the current

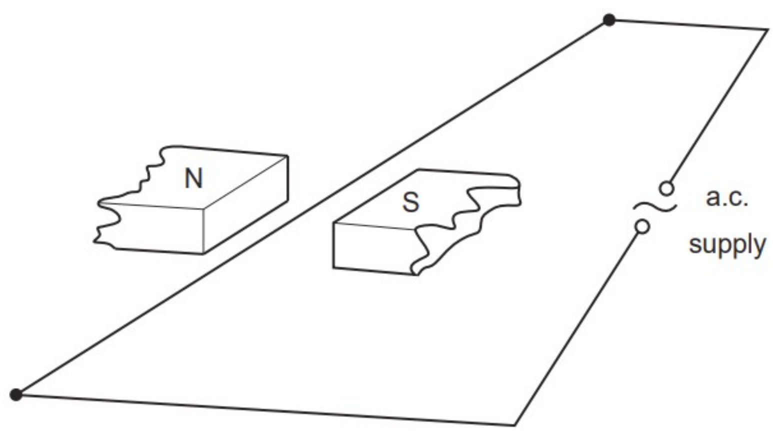
B to decrease the voltage and increase the current

C to increase the voltage and decrease the current

D to increase the voltage and increase the current

138. M/J/09/Q31

An a.c. supply is connected to a wire stretched between the poles of a magnet.



Which way will the wire move?

A left and right

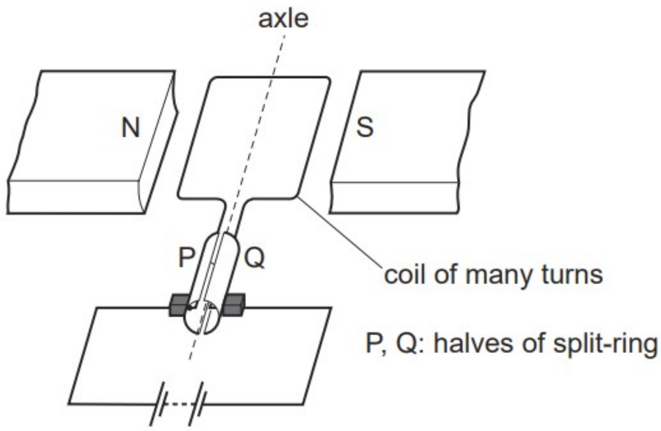
B right only

C up and down

D up only

139. M/J/09/Q32

A d.c. motor consists of a coil of many turns rotating in a fixed magnetic field. The coil is connected to a d.c. supply through a split-ring commutator.



Some changes are made, one at a time.

- The d.c. supply is reversed.
- The coil is turned before switching on, so that P starts on the right and Q on the left.
- The poles of the magnet are reversed.
- The turns on the coil are increased in number.

How many of these changes make the coil rotate in the opposite direction?

A 1 **B** 2 **C** 3 **D** 4

140. M/J/09/Q35

Why is electrical energy usually transmitted at high voltage?

A As little energy as possible is wasted in the transmission cables.

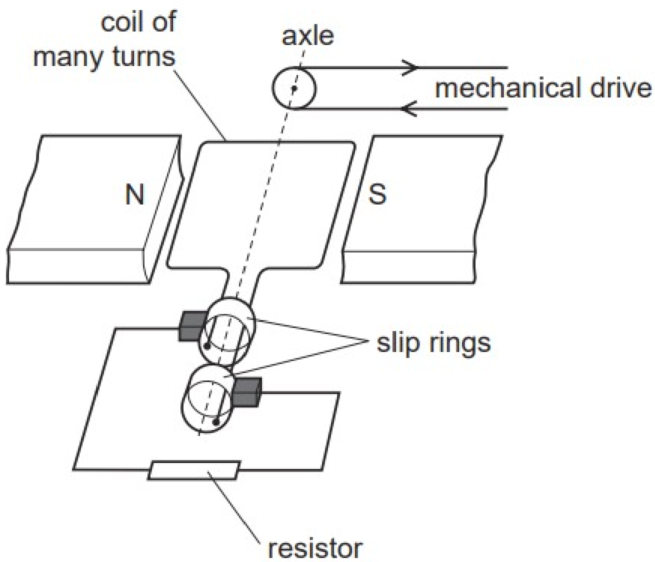
B The current in the transmission cables is as large as possible.

C The resistance of the transmission cables is as small as possible.

D The transmission system does not require transformers.

141. M/J/09/Q33

The diagram shows an a.c. generator connected to a resistor.



Some changes are made, one at a time.

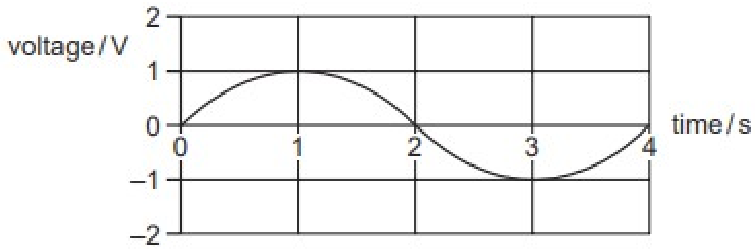
- The speed of the drive is changed.
- The strength of the magnets is changed.
- The number of turns in the coil is changed.
- The value of the resistor is changed.

How many of these alter the value of the e.m.f. generated in the coil?

A 1 **B** 2 **C** 3 **D** 4

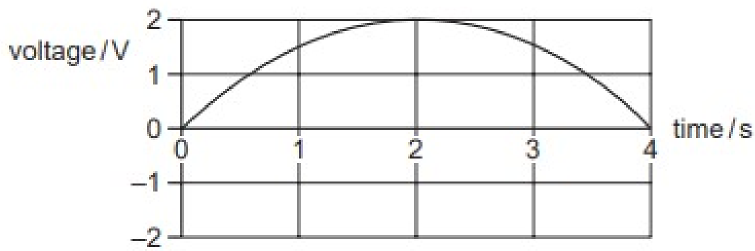
142. M/J/09/Q34

A simple a.c. generator produces a voltage that varies with time as shown.

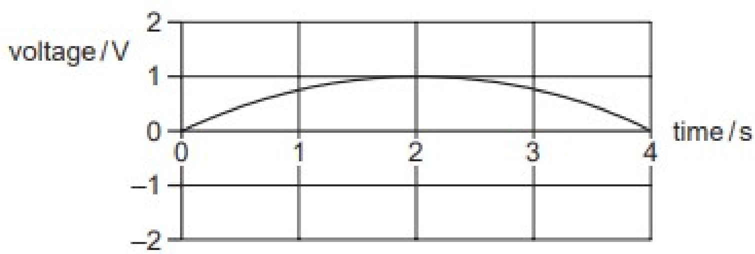


Which graph shows how the voltage varies with time when the generator rotates at twice the original speed?

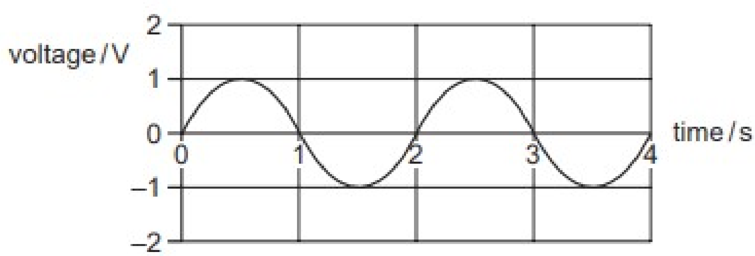
A



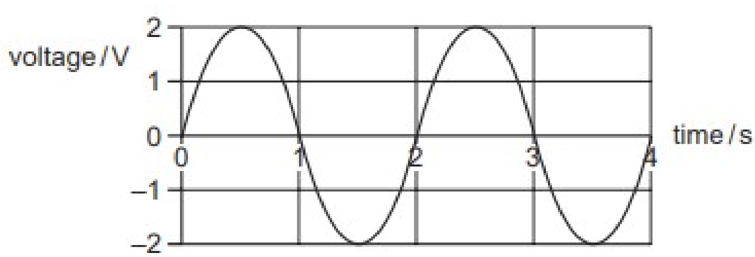
B



C

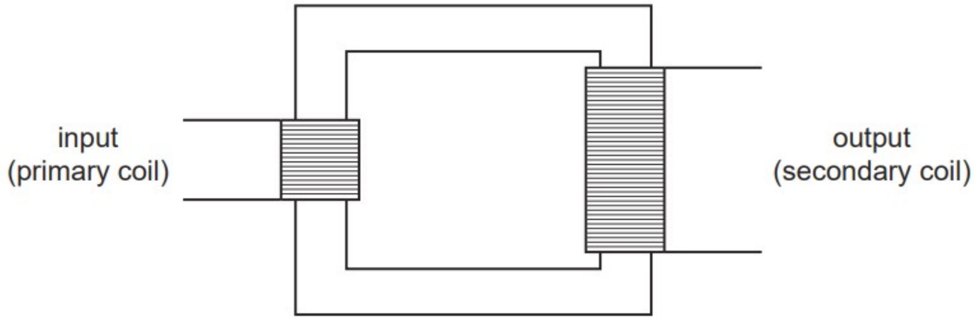


D



143. O/N/08/Q36

The diagram shows a working transformer.



Which statement is correct?

A The input voltage is d.c.

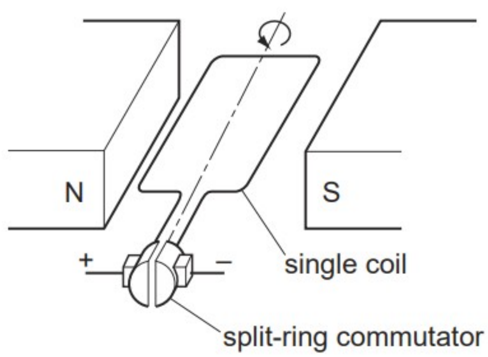
B The input voltage is greater than the output voltage.

C The input voltage is less than the output voltage.

D The input voltage is the same as the output voltage.

144. M/J/08/Q33

The diagram shows a simple electric motor.



The split-ring commutator reverses the current in the coil as the coil rotates.

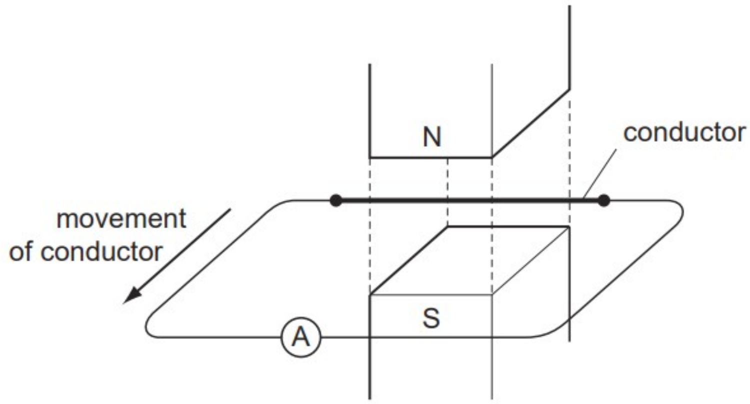
The coil is rotated 360° from the position shown.

How many times is the current in the coil reversed?

A 1 **B** 2 **C** 3 **D** 4

145. M/J/08/Q34

A conductor is moving horizontally across a vertical magnetic field.



An e.m.f. is induced in the conductor. No deflection is seen on the ammeter.

What is the reason for this?

A The ammeter is not between the poles.

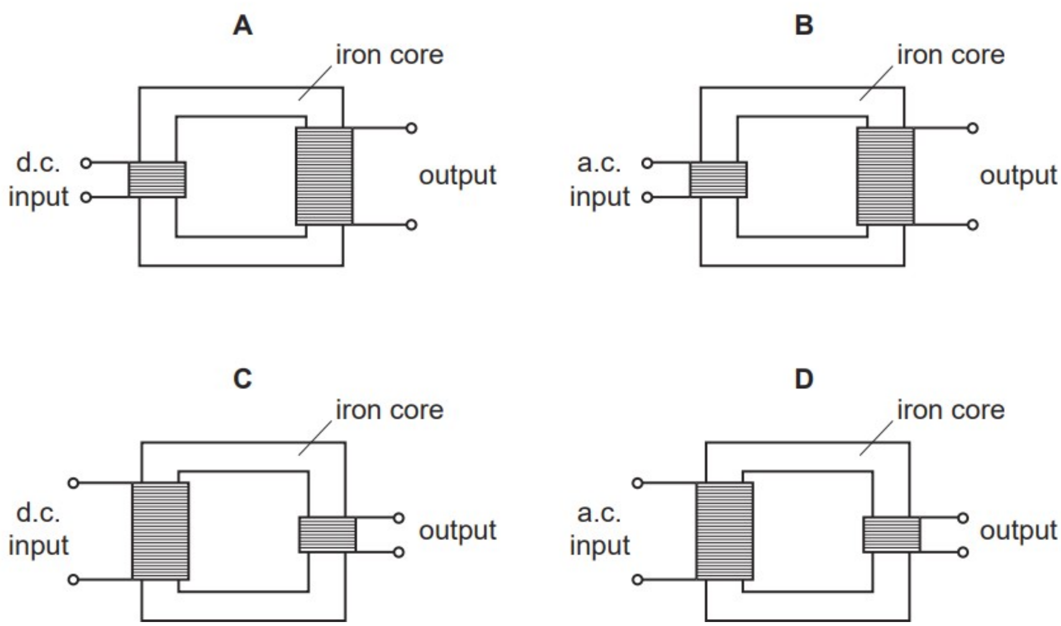
B The conductor is moving too slowly.

C The conductor is not cutting field lines.

D The poles are too close together.

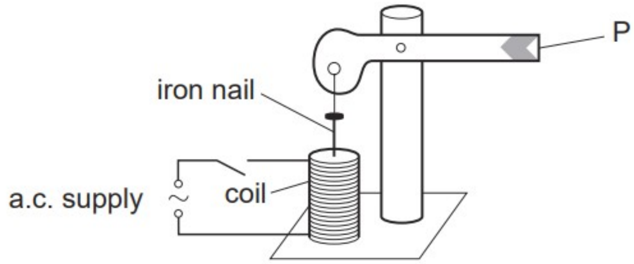
146. M/J/08/Q36

Which transformer arrangement produces an output voltage that is larger than the input voltage?



147. O/N/07/Q24

The diagram shows a model railway signal.



What does the end P do when the switch is closed?

A It goes down and stays down.

B It goes up and stays up.

C It goes down and then returns to its original position.

D It goes up and then returns to its original position.

148. O/N/07/Q25

Which part of a video tape recording system does **not** rely on magnetic material for its operation?

A the drive motor

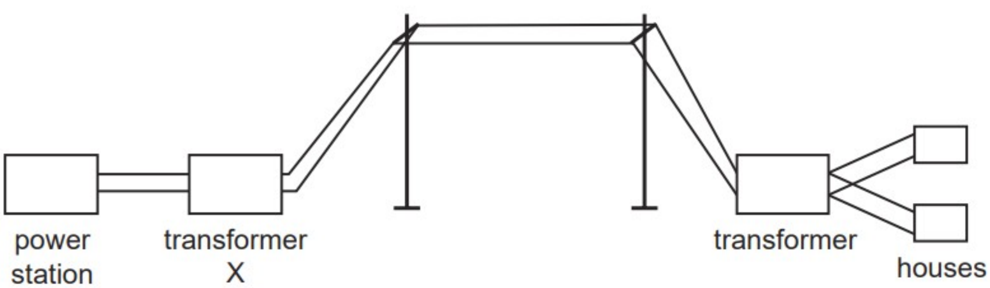
B the power lead

C the transformer

D the video tape

149. O/N/07/Q36

The diagram shows transformers and cables used to transmit electrical energy over long distances.



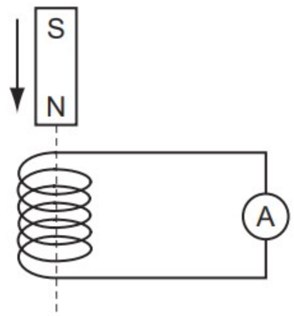
How does transformer X affect the voltage and the current from the power station?

	voltage	current
A	decreases it	decreases it
B	decreases it	increases it
C	increases it	decreases it
D	increases it	increases it

150. O/N/07/Q35

A small coil is connected to a sensitive ammeter. The ammeter needle can move to either side of the zero position.

As the magnet falls towards the coil, the ammeter needle moves quickly to the right of the zero position.



The magnet moves through the coil.

How does the ammeter needle move as the magnet falls away from the coil?

A It does not move.

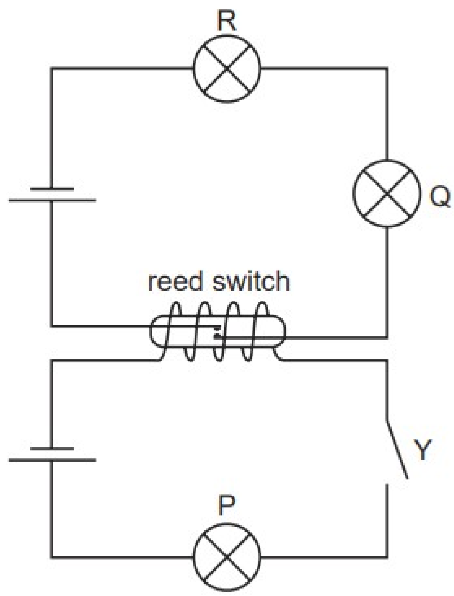
B It gives a steady reading to the right.

C It moves quickly to the left of the zero position and then returns to zero.

D It moves quickly to the right of the zero position and then returns to zero.

151. O/N/07/Q37

In the circuit shown, all lamps are identical and all cells are identical. The resistance of the coil of the reed switch is negligible.



One cell lights one lamp to normal brightness.

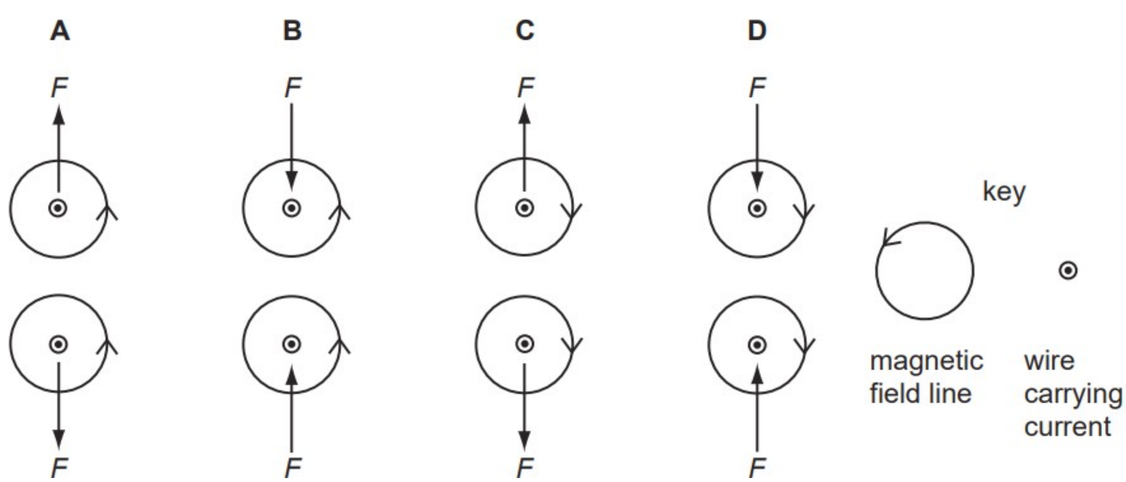
What is the brightness of the lamps when switch Y is closed?

	P	Q	R
A	dim	dim	dim
B	normal	dim	dim
C	normal	off	off
D	off	normal	normal

152. M/J/07/Q33

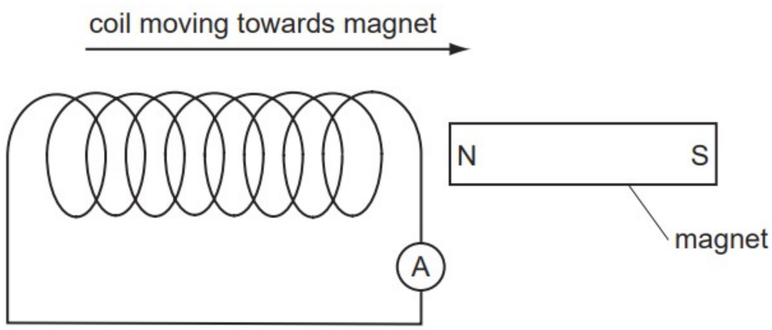
The diagrams show the forces *F* between two wires carrying currents out of the page. The magnetic fields close to the wires are also shown.

Which diagram is correct?



153. M/J/07/Q35

The diagram shows how a magnet and a coil may be used to induce an electric current.



How could the ammeter reading be increased?

A Move the coil more slowly.

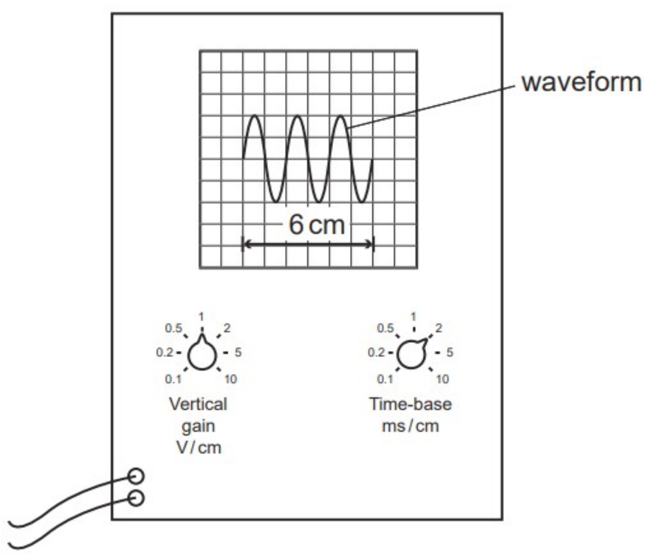
B Put a resistor in series with the ammeter.

C Turn the magnet round, then move the coil.

D Use a coil with more turns.

154. M/J/07/Q37

A waveform is displayed on a cathode-ray oscilloscope. The length of three cycles of the waveform is 6 cm. The vertical gain and the time-base settings are shown on the diagram.

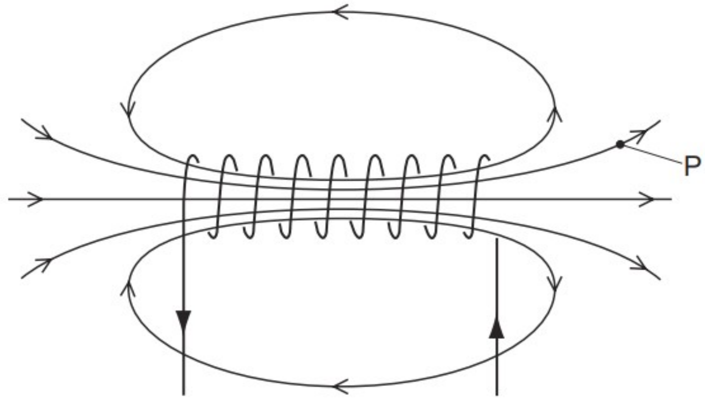


What is the time taken for one cycle of the waveform?

A 1 ms **B** 2 ms **C** 4 ms **D** 6 ms

155. O/N/06/Q24

A current in a solenoid creates a magnetic field.

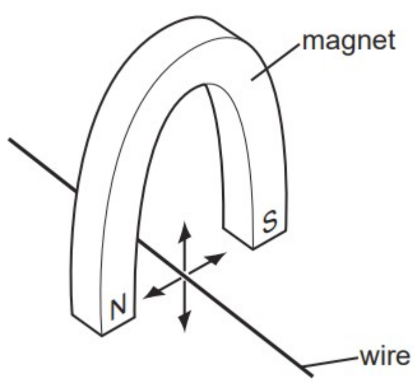


What is the effect on the magnetic field at the point P of using a larger current in the opposite direction?

	field strength	field direction
A	decreases	reverses
B	decreases	unchanged
C	increases	reverses
D	increases	unchanged

156. O/N/06/Q31

A copper wire is held between the poles of a magnet.



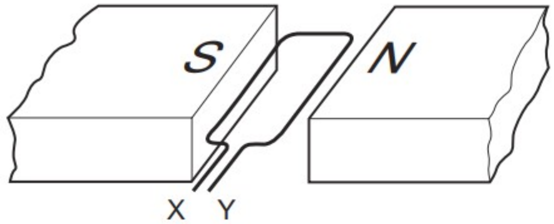
The current in the wire can be reversed. The poles of the magnet can also be changed over.

In how many of the four directions shown can the force act on the wire?

A 1 **B** 2 **C** 3 **D** 4

157. O/N/06/Q32

The diagram shows a coil in a magnetic field.



When the coil is part of a d.c. motor, what must be connected directly to X and Y?

A d.c. supply

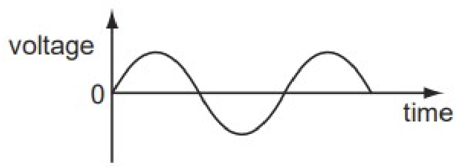
B slip rings

C soft-iron core

D split-ring commutator

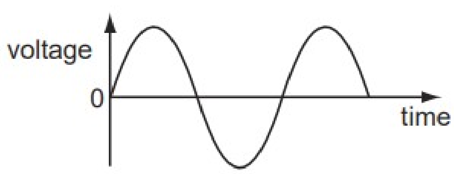
158. O/N/06/Q33

A transformer has more turns on the secondary coil than on the primary. The graph shows how the input voltage varies with time.

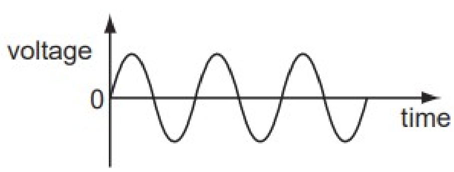


Which graph, drawn to the same scale as the input graph, shows how the output voltage varies with time?

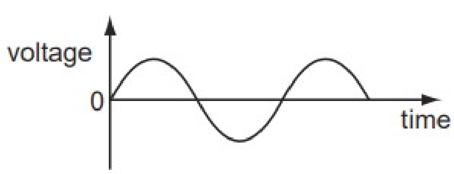
A



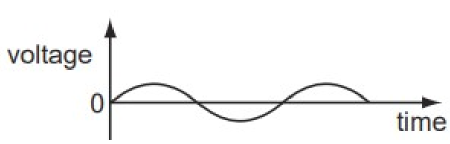
B



C



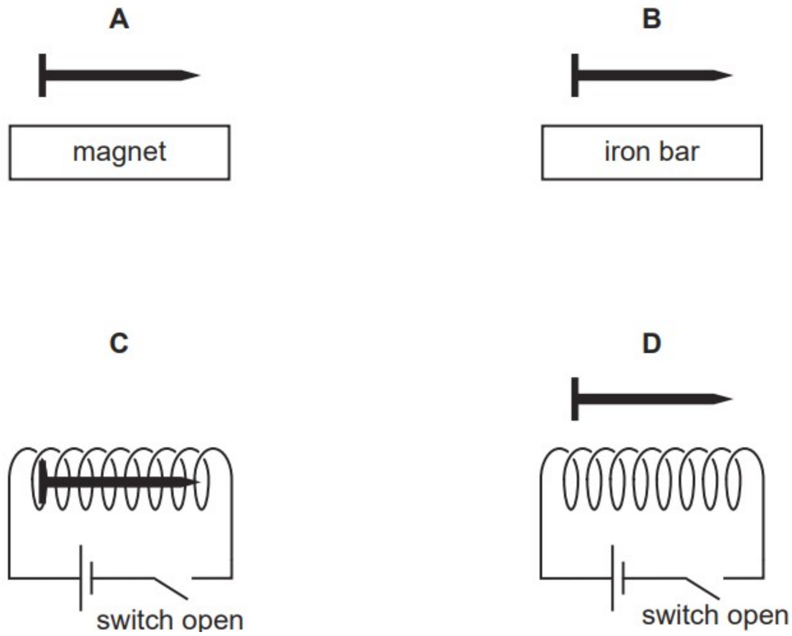
D



159. M/J/06/Q27

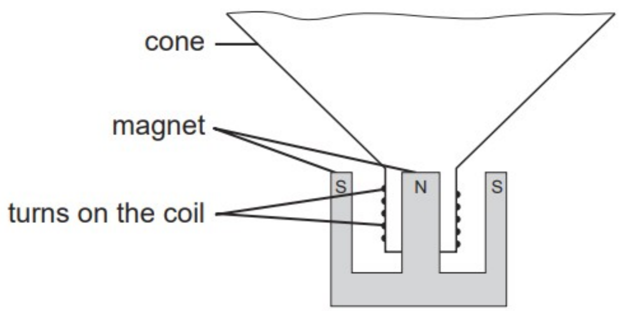
The diagrams show an iron nail in four different situations.

In which diagram will the nail become an induced magnet?



160. M/J/06/Q28

The diagram shows parts of a loudspeaker.



Which type of current is passed through the coil and why?

	current passed through coil	reason why
A	alternating	to keep the magnetic field constant
B	alternating	to make the coil vibrate
C	direct	to keep the magnetic field constant
D	direct	to make the coil vibrate

161. M/J/06/Q36

Why is a transformer used to connect a generator in a power station to a long distance transmission line?

A to decrease the voltage and decrease the current

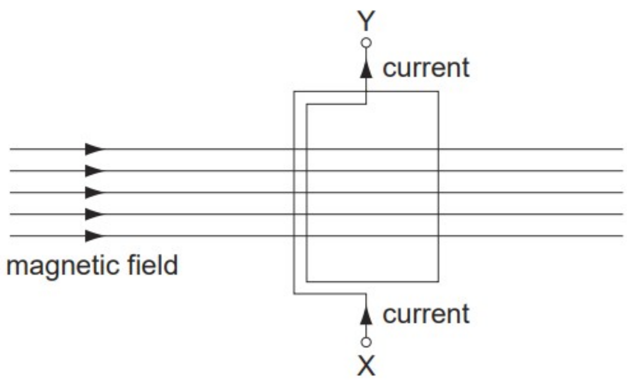
B to decrease the voltage and increase the current

C to increase the voltage and decrease the current

D to increase the voltage and increase the current

162. O/N/05/Q32

A coil, carrying a current, is arranged within a magnetic field. The coil experiences forces that can make it move.



In which direction does the coil move?

A along the magnetic field

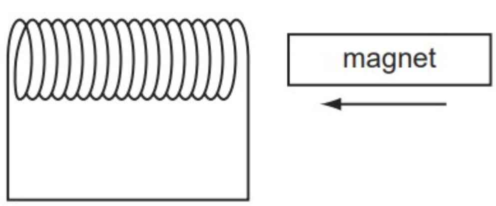
B from X to Y

C out of the paper

D turns about the axis XY

163. O/N/05/Q33

A magnet is pushed horizontally towards a coil of insulated wire, inducing an e.m.f. in the coil.



In which direction does the induced e.m.f. make the coil move?

A away from the magnet

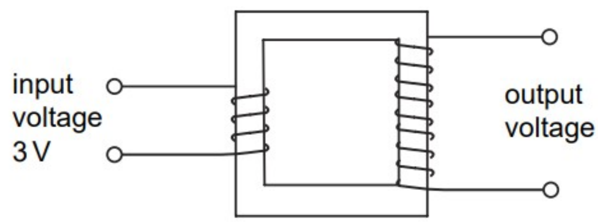
B towards the magnet

C downwards

D upwards

164. O/N/05/Q34

A step-up transformer with 100 % efficiency has an input voltage of 3 V and an input current of 2 A.

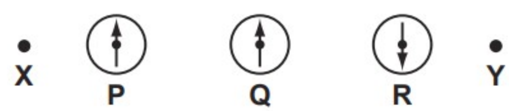


Under these conditions, what output voltage and output current could be obtained?

	output voltage / V	output current / A
A	1	6
B	2	3
C	4	1
D	6	1

165. M/J/05/Q32

X and **Y** are wires carrying electric currents at right angles to the page. **P**, **Q** and **R** are plotting compasses. Any effect of the Earth's magnetic field has been ignored.



What is true about the direction and size of the currents?

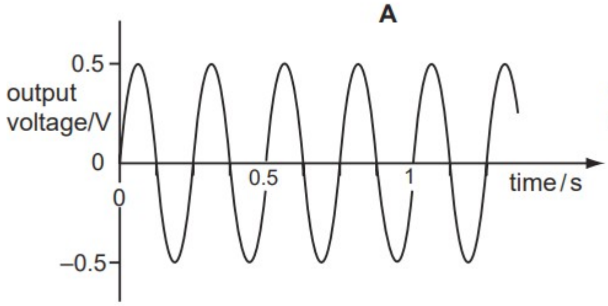
	direction of currents	size of currents
A	same	larger in **X** than in **Y**
B	same	smaller in **X** than in **Y**
C	different	larger in **X** than in **Y**
D	different	smaller in **X** than in **Y**

166. M/J/05/Q33

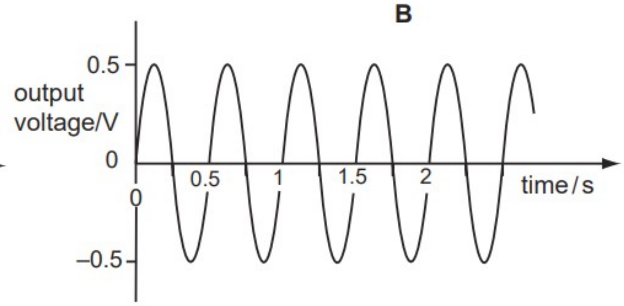
A girl turns the handle of a small a.c. generator four times each second. The generator produces a maximum output voltage of 0.5 V.

Which of the following graphs best shows this?

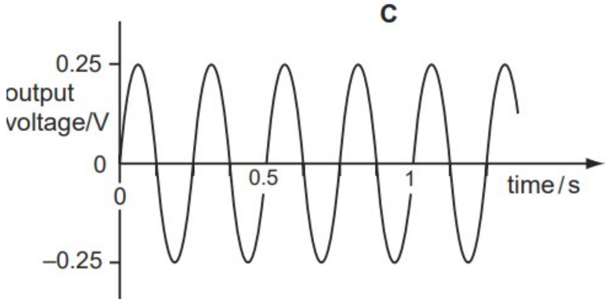
A



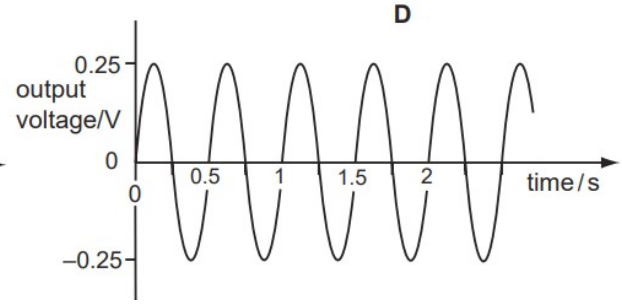
B



C



D



167. M/J/05/Q34

Which statement about the action of a transformer is correct?

A An e.m.f. is induced in the secondary coil when an alternating voltage is applied to the primary coil.

B An e.m.f. is induced in the secondary coil when there is a steady direct current in the primary coil.

C The current in the secondary coil is always larger than the current in the primary coil.

D The voltage in the secondary coil is always larger than the voltage in the primary coil.

168. M/J/05/Q35

Electrical energy is transmitted at high alternating voltages.

What is **not** a valid reason for doing this?

A At high voltage, a.c. is safer than d.c.

B For a given power, there is a lower current with a higher voltage.

C There is a smaller power loss at higher voltage and lower current.

D The transmission lines can be thinner with a lower current.